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Invited Review

Shock metamorphism of quartz in nature and experiment: II. Significance in geoscience♦

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Abstract–The occurrence of shock metamorphosed quartz is the most common petrographic criterion for the identification of terrestrial impact structures and lithologies. Its utility is due to its almost ubiquitous occurrence in terrestrial rocks, its overall stability and the fact that a variety of shock metamorphic effects, occurring over a range of shock pressures, have been well documented. These shock effects have been generally duplicated in shock recovery experiments and, thus, serve as shock pressure barometers. After reviewing the general character of shock effects in quartz, the differences between experimental and natural shock events and their potential effects on the shock metamorphism of quartz are explored. The short pulse lengths in experiments may account for the difficulty in synthesizing the high-pressure polymorphs, coesite and stishovite, compared to natural occurrences. In addition, post-shock thermal effects are possible in natural events, which can affect shock altered physical properties, such as refractive index, and cause annealing of shock damage and recrystallization. The orientations of planar microstructures, however, are unaffected by post-impact thermal events, except if quartz is recrystallized, and provide the best natural shock barometer in terms of utility and occurrence. The nature of planar microstructures, particularly planar deformation features (PDFs), is discussed in some detail and a scheme of variations in orientations with shock pressure is provided. The effect of post-impact events on PDFs is generally limited to annealing of the original glass lamellae to produce decorated PDFs, resulting from the exsolution of dissolved water during recrystallization. Basal (0001) PDFs differ from other PDF orientations in that they are multiple, mechanical Brazil twins, which are difficult to detect if not partially annealed and decorated. The occurrence and significance of shock metamorphosed quartz and its other phases (namely, coesite, stishovite, diaplectic glass and lechatelierite) are discussed for terrestrial impact structures in both crystalline (non-porous) and sedimentary (porous) targets. The bulk of past studies have dealt with crystalline targets, where variations in recorded shock pressure in quartz have been used to constrain aspects of the cratering process and to estimate crater dimensions at eroded structures. In sedimentary targets, the effect of pore space results in an inhomogeneous distribution in recorded shock pressure and temperature, which requires a different classification scheme for the variation of recorded shock compared to that in crystalline targets. This is discussed, along with examples of variations in the relative abundances of planar microstructures and their orientations, which are attributed to textural variations in sedimentary target rocks. Examples of the shock metamorphism of quartz in distal ejecta, such as at the K/T boundary, and from nuclear explosions are illustrated and are equivalent to that of known impact structures, except with respect to characteristics that are due to long-term, post-shock thermal effects. Finally, the differences between the deformation and phase transformation of quartz by shock and by endogenic, tectonic and volcanic processes are discussed. We confirm previous conclusions that they are completely dissimilar in character, due to the vastly different physical conditions and time scales typical for shock events, compared to tectonic and volcanic events. Well-characterized and documented shock effects in quartz are unequivocal indicators of impact in the natural environment.

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INTRODUCTION

Quartz is a rock-forming mineral in the crust of the Earth, Moon and Mars (if the SNC meteorites are derived from Mars, as is generally believed) and, very subordinately, in stony meteorites. Except for the Earth, however, quartz occurs as an accessory phase in all these bodies. In the case of the Earth, the occurrence of

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"shocked" quartz is the most common petrographic criterion for the identification of terrestrial impact structures. This is due to the clear diagnostic nature of shock metamorphic effects in quartz (Stöffler and Langenhorst, 1994), its relatively stable nature and its nearly ubiquitous occurrence in terrestrial crustal rocks. Part of the utility of quartz as a shock indicator is the occurrence of unequivocal shock effects over a wide pressure range. Approximately 80% of known terrestrial impact structures contain quartz with so-called planar deformation features. A much smaller percentage, ~18%, have documented occurrences of the coesite and/or stishovite high-pressure polymorphs. This is due, in part, to the metastable nature of the polymorphs, particularly with respect to immediate and secondary post-shock thermal events (Dachille *et al.*, 1963; Skinner and Fahey, 1963) and, in part, to the relatively laborious procedures required to concentrate and identify the polymorphs (Fahey, 1964; Stöffler, 1971a; McHone *et al.*, 1989).

Shocked quartz has been found only in two terrestrial geological settings: (1) in direct spatial relation to impact structures and (2) as a constituent of distal or global ejecta deposits. In this contribution, we complete the review initiated by Stöffler and Langenhorst (1994), which focussed largely on the fundamental observations and theory relating to the shock metamorphism of quartz. Here, after a brief review of the general characteristics of shock effects in quartz (for details, see Stöffler and Langenhorst, 1994), we focus on the occurrence and character of shocked quartz in the natural environment and its relation to shock recovery experimental data. We also illustrate the utility of studies of the shock metamorphism of quartz in constraining impact processes and the observed differences that exist in the response of quartz to shock in crystalline and porous sedimentary targets.

GENERAL CHARACTERISTICS OF SHOCK EFFECTS IN QUARTZ

Quartz subjected to shock pressures >5 GPa displays various types of residual shock effects, which vary with increasing shock pressure and depend, to some degree, on the pre-shock temperature (Fig. 1). Quartz starts to melt under shock compression at >50 GPa and vaporizes at >100 GPa. Within the pressure range of ~5 GPa to >50 GPa, the following major types of shock effects are observed: (1) mosaicism, (2) planar microstructures (planar fractures and planar deformation features), (3) partial to complete isotropization or amorphization (diaplectic glass), (4) partial transformation to stishovite and coesite and (5) melting and quenching to lechatelierite.

Mosaicism

Under the microscope, shock-induced mosaicism appears as a highly irregular "mottled" extinction pattern in quartz with or without planar microstructures. In single crystal x-ray diffraction patterns, it causes streakiness and broadening of the diffraction spots. Domain sizes are >200 nm. Domain-size distribution is distinctly heterogeneous, but domain size decreases with increasing pressure. Attempts to derive a quantitative correlation between shock pressure and degree of mosaicism indicate that the domain size and the internal strain, derived from x-ray diffraction, provide the best measure of recorded shock pressure (Dachille *et al.*, 1968; Hörz and Quaide, 1973; Hanss *et al.*, 1978; Schneider *et al.*, 1984; Ashworth and Schneider, 1985; Langenhorst, 1994; Langenhorst and Deutsch, 1994).

Planar Microstructures

Quartz fails mechanically upon shock compression and release in two basic modes: (1) by the formation of irregular fractures, which are not diagnostic shock effects and (2) by the formation of regular planar microstructures, which are diagnostic shock effects and are oriented parallel to rational crystallographic planes of low Miller indices. The term "planar microstructures" (Stöffler and Langenhorst, 1994) is used as a collective term for two types of microstructures: planar fractures (PFs) and planar deformation features (PDFs). The PDFs occur as either so-called non-decorated or decorated PDFs with the latter being formed by annealing of non-decorated PDFs.

Planar fractures usually form sets of parallel open fissures, with a spacing of ~>20 μm and are oriented along crystallographic planes, such as (0001) and $\{10\bar{1}1\}$ (Fig. 2). They commonly act as domain boundaries for the much more closely spaced (2 μm–10

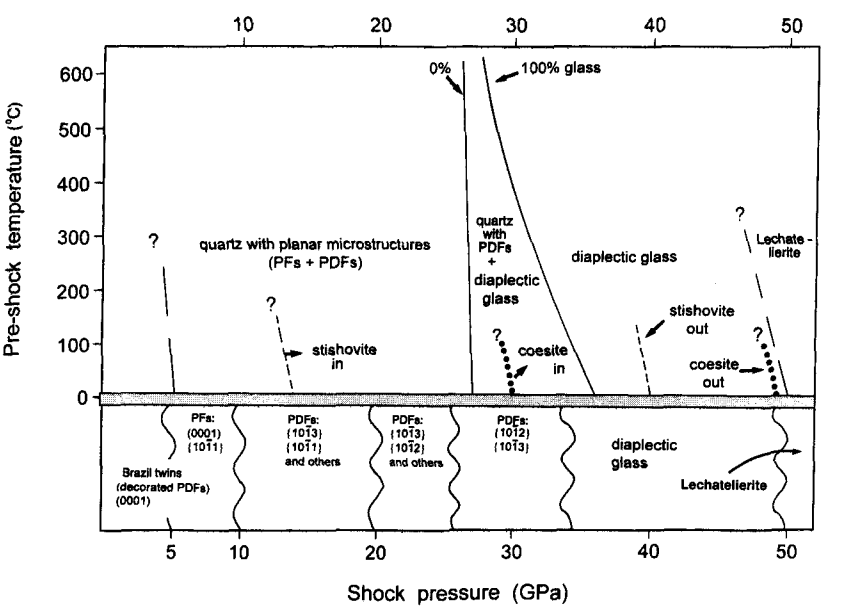


FIG. 1. Occurrence of specific shock metamorphic effects in quartz from crystalline (non- porous) targets as a function of shock pressure and pre-shock temperature. Lower panel indicates orientations of most common planar fractures (PFs) and planar deformation features (PDFs) in specific pressure ranges. See Table 1 for other PDF orientations and information on variations in accompanying optical properties of quartz.

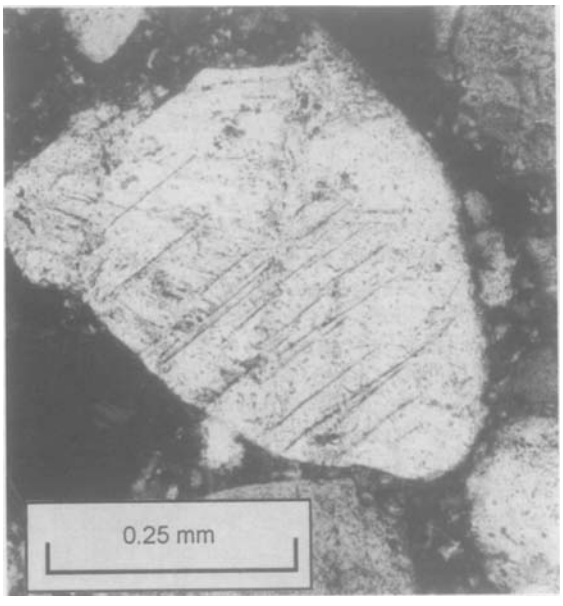


FIG. 2. Photomicrograph (crossed polars) of planar fractures in quartz in a breccia from the Aorounga structure.

μm) PDFs. The PDFs occur as multiple sets of parallel, planar optical discontinuities, which are sometimes resolvable by the optical microscope as thin lamellae (Fig. 3) or as planes decorated with tiny vugs ("decorated PDFs"; Fig. 3). Up to 15 sets of PDFs of different orientation can be present in a single quartz crystal. The most frequent orientations are: {10$\bar{1}$3}, {10$\bar{1}$2}, {10$\bar{1}$1}, (0001), {11$\bar{2}$1}, {11$\bar{2}$2}, {21$\bar{3}$1}, {22$\bar{4}$1}, {51$\bar{6}$1}, {10$\bar{1}$0} and {11$\bar{2}$0}, with some other rare orientations (see Tables 3–5 in Stöffler and Langenhorst, 1994). There are two fundamentally different structural types of PDFs. When fresh, most PDFs are amorphous lamellae 30 nm to 200 nm thick. The PDFs parallel to (0001), however, are multiple mechanical Brazil twin lamellae (Goltrant *et al.*, 1991, 1992; Leroux *et al.*, 1994). This was noted in an early TEM study (Kieffer *et al.*, 1976) and Carter (1965, 1968) in early optical studies discriminated between the character of basal and non-basal orientations of PDFs.

These basal PDFs cannot be identified easily by the optical microscope, unless they are decorated PDFs. This has considerable consequences for the observational data base on PDFs in quartz. For example, to date, such basal mechanical twins have not been identified in experimentally shocked quartz. There may be two reasons for this. Earlier experimental samples have only been studied optically, and the PDFs were not decorated (*e.g.*, Hörz, 1968; Müller and Défourneaux, 1968). In addition, basal twins form only in the low-shock pressure range (<10 GPa), where there is a general lack of shock recovery experiments. Static high-pressure experiments indicate that mechanical basal Brazil twins can only form under rapidly applied and high shear stresses >3 GPa–4 GPa (McLaren *et al.*, 1967). They are well-documented at impact structures with other shock effects (*e.g.*, other PDF orientations, diaplectic glasses, high-pressure polymorphs, *etc.*), and they are considered to be diagnostic indicators for impact. Stöffler and Langenhorst (1994) combined critical orientations of planar microstructures and refractive index data in order to derive a system of progressive stages of shock metamorphism in quartz (Fig. 1; Table 1), which has been calibrated experimentally.

Diaplectic Glass

The refractive indices and the birefringence of quartz decrease gradually above a certain degree of shock, which is characterized by the coexistence of PDFs parallel to {10$\bar{1}$3} and {10$\bar{1}$2} (Table 1),

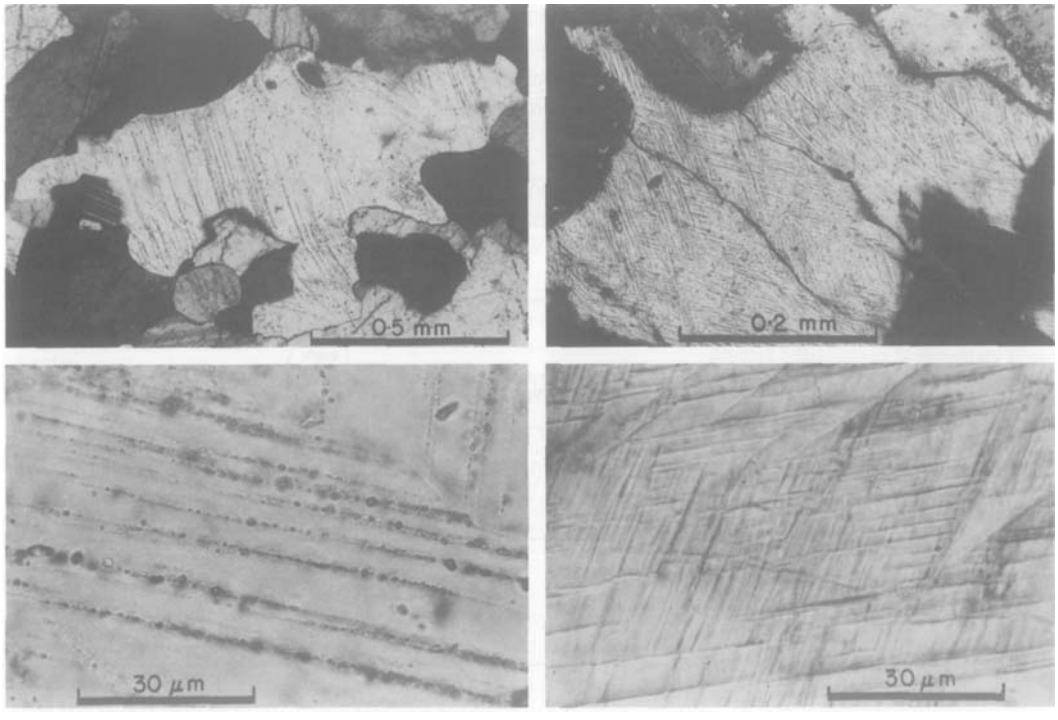


FIG. 3. (left) Photomicrograph (crossed polars) of nondecorated (fresh) planar deformation features in quartz from the Mistastin structure. (right) Photomicrograph (crossed polars) of decorated (annealed) planar deformation features in quartz from the Charlevoix structure. See text for further descriptive details.

TABLE 1. Progressive stages of shock metamorphism of quartz (in non-porous rocks), modified from Stöffler and Langenhorst (1994) and based on data in Engelhardt *et al.* (1968), Robertson *et al.* (1968), Engelhardt and Bertsch (1969), Stöffler (1971a,b), Stöffler and Hornemann (1972), Robertson (1975) and Robertson and Grieve (1977).

Shock stage	Critical sets of PFs and PDFs*	Additional sets of PFs and PDFs	Optical properties of quartz
(1) Very weakly shocked	PFs: PDFs: (0001)	PFs: {10$\bar{1}$1} rare PDFs: none	Normal
(2) Weakly shocked	PDFs: {10$\bar{1}$3}	PFs: {10$\bar{1}$1}, (0001) PDFs: (0001)	Normal
(3) Moderately shocked	PDFs: {10$\bar{1}$3}	PFs: {10$\bar{1}$1}, (0001) rare PDFs: {11$\bar{2}$1}, {11$\bar{2}$2}, (0001), {10$\bar{1}$0} + {11$\bar{2}$1}, {10$\bar{1}$1}, {21$\bar{3}$1}, {51$\bar{6}$1}	Normal or slightly reduced refractive indices
(4) Strongly shocked	PDFs: {10$\bar{1}$2}, {10$\bar{1}$3}	PFs: rare or absent PDFs: {11$\bar{2}$1}, {11$\bar{2}$2}, (0001), {10$\bar{1}$0} + {11$\bar{2}$1}, {10$\bar{1}$1}, {21$\bar{3}$1}, {51$\bar{6}$1}	Reduced refractive indices: 1.546–1.480
(5) Very strongly shocked	PDFs: {10$\bar{1}$2}, {10$\bar{1}$3}	rare	Reduced refractive indices: <1.480
(6) Extremely shocked			Diaplectic glass, refractive index: 1.468–1.461
(7) Shock fused			Lechatelierite, refractive index: 1.460–1.458

Additional PDF orientations in stages (3) and (4): {22$\bar{4}$1}, {40$\bar{4}$1}, {51$\bar{6}$0} (Robertson *et al.*, 1968), and {33$\bar{4}$1} (Engelhardt *et al.*, 1968).
* See text for nature of PFs and PDFs.

until it becomes isotropic and the state of diaplectic glass is achieved (Stöffler and Langenhorst, 1994). The diaplectic glass has the same morphology as the original quartz crystal and shows no evidence of fluid textures (Fig. 4). The refractive index of diaplectic quartz glass is 1.468–1.461, compared to 1.459 for synthetic silica glass, and continues to decrease with increasing pressure until lechatelierite (shock-fused quartz), with a refractive index of 1.460–1.458, is formed (Fig. 1; Table 1).

Coesite and Stishovite

The high-pressure polymorphs, coesite and stishovite, occur as very fine-grained aggregates (Fig. 5) that are formed by partial transformation of the host quartz. The shock-pressure range over which these phases crystallize and their textural settings are different in crystalline rocks and in porous sedimentary rocks (*e.g.*, Kieffer, 1971; Stöffler, 1971a; Kieffer *et al.*, 1976). In crystalline or dense rocks, coesite is found in quartz with PDFs and strongly lowered refractive index and, more commonly, in diaplectic glass; whereas stishovite occurs most commonly in quartz with PDFs and less frequently in diaplectic glass (Stöffler, 1971a). The estimated pressure ranges for the formation of coesite and stishovite are 30 GPa–60 GPa and 12 GPa–45 GPa, respectively (Stöffler and Langenhorst, 1994).

In porous sandstone (Kieffer, 1971; Kieffer *et al.*, 1976), because of the very inhomogeneous pressure (and temperature) distribution on the scale of individual quartz grains, coesite is formed at relatively low-average shock pressure (estimated at 5 GPa–10 GPa) and coexists with >80% of quartz displaying planar fractures and diaplectic quartz glass. At medium shock levels (~13 GPa–30 GPa), an assemblage of fractured quartz, amorphous quartz, coesite and stishovite is observed. At higher degrees of shock (> 30 GPa), the main assemblage is lechatelierite and coesite, which forms a highly vesiculated rock.

Lechatelierite

Lechatelierite is the product of the highest degree of shock (Fig. 1; Table 1) and is commonly observed in impact craters formed in sedimentary rocks or unconsolidated sediments (French and Short, 1968; Kieffer *et al.*, 1976; Hörz *et al.*, 1989; Stöffler and Langenhorst, 1994). There it forms highly vesiculated glass with flow structures (Fig. 6), observed frequently with less shocked crystalline quartz and coesite. In impact craters in non-porous, crystalline targets, lechatelierite is found only in impact melt glasses and rocks in the form of inclusions and schlieren (*e.g.*, Engelhardt, 1972), which result from "secondary" melting of highly shocked quartz that was mixed into the impact melt during the crater-forming process. In TEM observations of experimentally and naturally shocked quartz, lechatelierite occurs as submicroscopic veins or fillings of cracks within quartz displaying PDFs (Gratz *et al.*, 1988a,b, 1992a; Martinez *et al.*, 1993). These veins appear to be the result of localized strain and temperature peaks that are caused by shock wave reverberations.

COMPARISON OF EXPERIMENTAL AND NATURAL SHOCK EFFECTS

Differences between natural and experimental shock effects can be appreciated by comparing shock wave profiles and post-shock histories. This also permits an assessment of which experimentally calibrated shock effects are most reliable in estimating and reconstructing the physical conditions in a natural impact event.

Shock Wave Profiles

Simplified shock profiles in nature and experiments are depicted in Fig. 7. Complexities in the shock wave structure, which can result, for example, from phase transitions, are neglected. Shock wave profiles are generally characterized by three regimes: (1) the loading phase, (2) the compression phase and (3) the unloading or decompression phase.

Independent of the type of projectile accelerating system (gas gun or high explosives), shock pressure loading is achieved by two

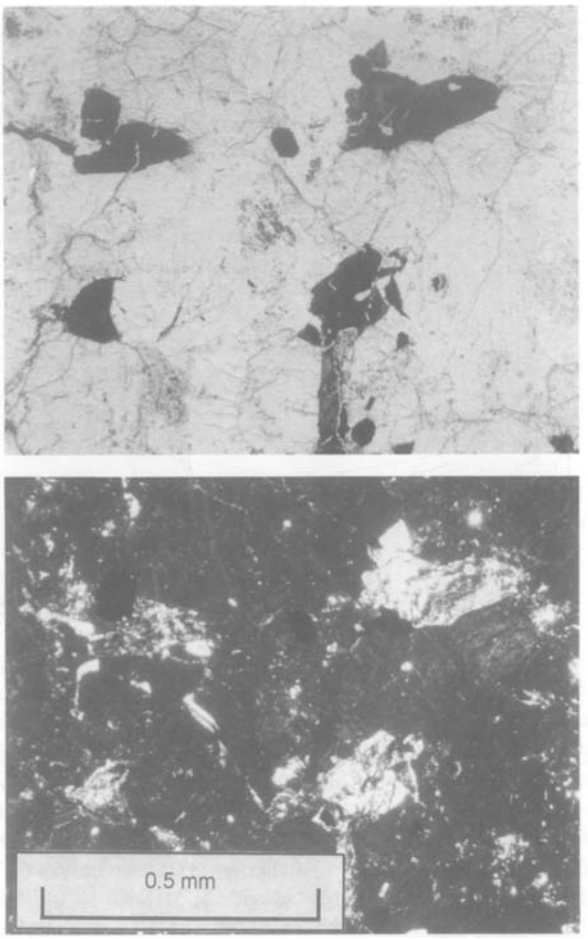


FIG. 4. Photomicrograph of diaplectic quartz and feldspar (maskelynite) glass in a microgranite from the Mistastin structure. (top) Plane polarized light. Rock consists of quartz (rounded, high relief, colorless), feldspar (low relief, colorless), biotite (dark) and opaques (dark). (bottom) Crossed polars. All quartz and feldspar are diaplectic glass (isotropic) while biotite retains birefringence.

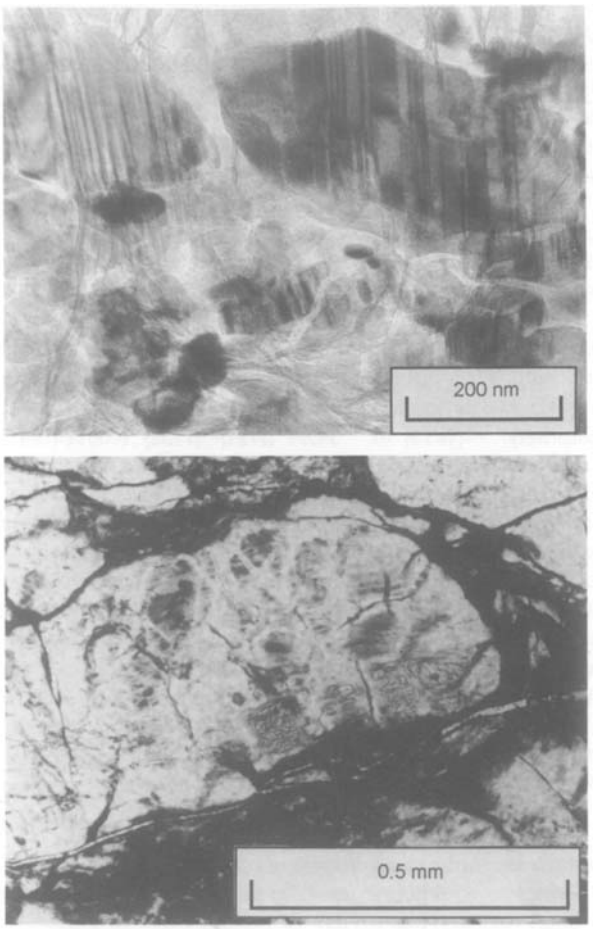


FIG. 5. (top) Transmission electron microscope (TEM) bright field image of coesite grains (dark) with surrounding montmorillonite from the Ries structure. Coesite shows polysynthetic twinning parallel to (010). (bottom) Photomicrograph (plane polarized light) of stishovite (mottled dark grey) in quartz (white) in pseudotachylite (black) at the Vredefort structure. Photograph supplied by J. C. White.

basic experimental techniques: impedance and reverberation (Fig.7a). In the impedance technique, a single shock wave is generated and decays within the sample to ambient pressure before reflections can take place at sample-container interfaces. By contrast, the maximum pressure in reverberation experiments is produced in steps by multiple reflections at these interfaces (see Fig. 10 in Stöffler and Langenhorst, 1994). If single crystals are shocked, the reverberation technique may better simulate the natural case, as reflections between the different rock constituents are typical for a naturally shocked sample. The shock impedance differences, however, between container (metal) and single crystal (silicate) in reverberation experiments are greater than those between the different minerals in a rock. Therefore, the pressure jumps during loading in reverberation experiments are larger than in nature. If a polycrystalline sample is shocked with the impedance technique, shock wave reflections also occur between the individual grains. The disadvantage, however, of the impedance technique is the lower pressure achieved compared to the reverberation technique. There is little known about the influence of experimental design on the production of shock effects. Langenhorst and Deutsch (1994) have suggested that shock effects in quartz are more homogeneously distributed when the reverberation technique is used. Since the loading time (ns) and strain rates ($\sim 10^6$ s^{-1}–10^9 s^{-1}) are comparable in nature and experiment, the minor differences in the loading paths are not sufficient to restrict the application of experimentally derived data to naturally shocked rocks.

The duration of the compression and decompression phases in experiments, however, differ considerably from those in nature. The pulse length in experiments is typically ~1 μs; whereas, large natural impacts, such as the formation of the Sudbury, Vredefort or Chicxulub structures, can have pressure durations of up to seconds (Fig. 7a,b). (The principal characteristics of the terrestrial impact structures mentioned in the text are given in Table 2.) In the decompression phase, the high-pressure shock state is released to ambient pressure by a rarefaction wave. The duration of the decompression phase significantly increases with progressive propagation of the shock wave (Fig. 7c). In theory, the time for decompression should be of the same order as the original pulse length (Boslough and Asay, 1993). This means, for the natural case, that the target rocks in the floors of impact structures experience a much longer decompression phase than target rocks near the surface.

The differences in the time scales of experimental and natural shock events result in some documented differences in recorded shock effects. For example, the extensive formation of stishovite

and coesite only occurs in nature. Up to 40% of the original quartz can be replaced by coesite and up to 3% by stishovite in natural impact events (Kieffer, 1971; Stöffler, 1971a). Synthetic coesite has been reported, however, only in one shock recovery experiment (Deribas *et al.*, 1966), which has not been duplicated. Stishovite has been produced in several shock recovery experiments but in extremely small quantities (<<1%; Kleemann and Ahrens, 1973; Ashworth and Schneider, 1985). These observations are in agreement with the hypothesis that the high-pressure phases are formed by rapid crystallization from a high-pressure melt (Langenhorst, 1994; Stöffler and Langenhorst, 1994) with the longer times in nature resulting in greater amounts of high-pressure phases. Despite the enormous differences in time scale, up to 6 orders of magnitude, other shock effects of quartz, such as planar microstructures, mosaicism, diaplectic glass and lechatelierite, have been experimentally reproduced to the same extent as in nature (Stöffler and Langenhorst, 1994 and references therein). It is, therefore, particularly these characteristics and diagnostic effects that can be used for the recognition of natural shock (impact) events and for shock barometry.

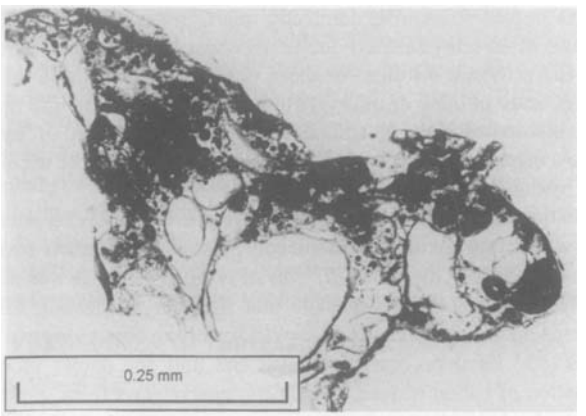


FIG. 6. Photomicrograph (plane polarized light) of highly vesiculated lechatelierite from suevite, Ries structure.

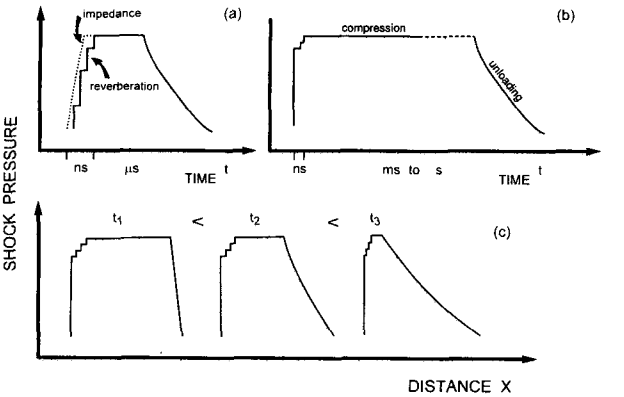


FIG. 7. Schematic of shock pressure as a function of time. (a) Experimental pressure-time path with shock loading by either reverberation or impedance occurring in nanoseconds, followed by compression and unloading over microseconds. (b) Natural pressure-time path in an impact event. Note that the loading time scales are similar to experiments, but the compression and unloading time scales are much longer, up to seconds in very large impact events. (c) Variation in pressure-time path in natural impact events with increasing time ($t_3 > t_2 > t_1$) or increasing distance of shock wave propagation from the point of impact. Note the shorter compression and longer unloading time at greater time/distance into the target. See text for more details.

TABLE 2. Principle attributes of impact structures referred to in the text.

Name	Country	Latitude	Longitude	Diameter* (km)	Age* (Ma)
Acraman	Australia	S 32° 01'	E 135° 27'	90	<450
Aorounga	Chad	N 19° 6'	E 19° 15'	12.6	<0.004
Avak	U.S.A.	N 71° 15'	W 156° 38'	12	>95
Azuara	Spain	N 41° 10'	W 0° 55'	30	<130
B.P. Structure	Libya	N 25° 19'	E 24° 20'	2.8	<120
Barringer	U.S.A.	N 35° 02'	W 111° 01'	1.2	0.049 ± 0.003
Brent	Canada	N 46° 05'	W 78° 29'	3.8	450 ± 30
Charlevoix	Canada	N 47° 32'	W 70° 18'	54	357 ± 15
Chicxulub	Mexico	N 21° 20'	W 89° 30'	180	64.98 ± 0.05
Chilyi	Kazakhstan	N 49° 10'	E 57° 51'	5.5	46 ± 7
Clearwater, West	Canada	N 56° 13'	W 74° 30'	36	290 ± 20
Crooked Creek	U.S.A.	N 37° 50'	W 91° 23'	7	320 ± 80
Gosses Bluff	Australia	S 23° 50'	E 132° 19'	22	142.5 ± 0.5
Janisjärvi	Russia	N 61° 58'	E 30° 55'	14	698 ± 22
Kamensk	Russia	N 48° 20'	E 40° 15'	25	65 ± 2
Kara	Russia	N 69° 12'	E 65° 00'	65	73 ± 3
Manicouagan	Canada	N 51° 23'	W 68° 42'	100	214 ± 1
Mien	Sweden	N 56° 25'	E 14° 52'	9	121.0 ± 2.3
Mistastin	Canada	N 55° 53'	W 63° 18'	28	38 ± 4
Oasis	Libya	N 24° 35'	E 24° 24'	11.5	<120
Popigai	Russia	N 71° 30'	E 111° 00'	100	35 ± 5
Puchezh-Katunki	Russia	N 57° 06'	E 43° 35'	80	220 ± 10
Ries	Germany	N 48° 53'	E 10° 37'	24	15.1 ± 1.0
Rochechouart	France	N 45° 50'	E 0° 56'	23	186 ± 8
Sierra Madera	U.S.A.	N 30° 36'	W 102° 55'	13	<100
Slate Islands	Canada	N 48° 40'	W 87° 00'	30	<350
Steinheim	Germany	N 48° 02'	E 10° 04'	3.8	15.1 ± 1.0
Sudbury	Canada	N 46° 36'	W 81° 11'	~250	1850 ± 3
Tookoonooka	Australia	S 27° 00'	E 143° 00'	55	128 ± 5
Vredefort	South Africa	S 27° 00'	E 27° 30'	~300	2016 ± 10
Wabar	Saudi Arabia	N 21° 30'	E 50° 28'	0.1	0.006 ± 0.002
Wanapitei	Canada	N 46° 45'	W 80° 45'	7.5	37 ± 2
West Hawk	Canada	N 49° 46'	W 95° 11'	2.44	100 ± 50

* Several buried or eroded impact structures have poorly constrained diameter estimates. Chicxulub, which is buried, however, represents an exceptional case, with other diameter estimates ranging as high as 300 km (Sharpton *et al.*, 1993). Age estimates are also of variable accuracy.

Post-Shock Thermal History

Natural and experimental shock effects can also differ as consequence of post-shock annealing and alteration, which are lacking in shock experiments. The diversity in the appearance of planar microstructures in naturally shocked quartz reflects the importance of post-shock history. An equivalent diversity in the appearance of planar microstructures is not observed in shock experiments. Investigations by TEM on experimentally shocked quartz have shown that fresh, non-decorated (at the optical scale) PDFs consist of amorphous silica lamellae (see Fig. 18 in Stöffler and Langenhorst, 1994), with the

exception of the (0001) Brazil twins (Müller, 1969; Ashworth and Schneider, 1985; Gratz *et al.*, 1988a, 1992a; Langenhorst, 1994). Non-decorated (at the optical scale) PDFs occur only in rapidly quenched samples from relatively young impact structures, in ejecta or in the products of shock recovery experiments and nuclear explosions. Decorated PDFs are secondary features, which are formed by annealing and aqueous alteration of non-decorated amorphous PDFs, and are the most common type of PDF occurring at terrestrial impact structures. The basal (0001) PDFs, as twins, can only be erased by melting or whole-scale recrystallization by metamorphism.

Recent TEM investigations have resulted in a clearer picture on how PDFs are modified by post-shock annealing and aqueous alteration (Goltrant *et al.*, 1991, 1992; Leroux and Doukhan, 1995). The solubility of water in amorphous and crystalline silica is different. Up to 6 wt% H_2O are soluble in silica glass (McMillan and Remmele, 1986); whereas, <100 ppm are soluble in quartz (Cordier and Doukhan, 1989). Following an impact event, water circulation in the brecciated and fractured impact lithologies is a common process. Therefore, aqueous fluids can be expected to dissolve in the amorphous PDFs (Fig. 8a). Subsequent annealing, which itself is assisted by the presence of water, initiates the recrystallization of the amorphous material in the PDFs. This leads initially to a mixture of amorphous silica and minute quartz crystallites (<10 nm; Fig. 8b), as observed by Goltrant *et al.* (1991, 1992). Progressive recrystallization causes coarsening of the quartz crystallites and, more importantly, exsolution of water within the former amorphous PDFs. The exsolution, in the form of bubbles, results from the newly formed quartz crystallites now being supersaturated in water. The pressure inside the bubbles increases with their increasing size. Relaxation of the overpressure is achieved by incorporating SiO_2, from the bubble surface, as extra half planes into the surrounding quartz lattice. A high density of sessile dislocation loops is formed, decorating the originally amorphous PDFs with the bubbles (Fig. 8c; McLaren *et al.*, 1983; Cordier and Doukhan, 1989). Such PDFs, containing a high density of dislocations and bubbles, are observed by TEM in quartz grains from K/T sites and impact structures younger than ~100 Ma (Goltrant *et al.*, 1991, 1992; Leroux and Doukhan, 1995). If annealing has progressed further, as is commonly the case in nature, the only remnants of PDFs are aligned bubbles, which still show the original orientation of the former amorphous lamellae (Fig. 8d). The dislocations have been transported out of the crystal or are reorganized in the form of subgrain boundaries.

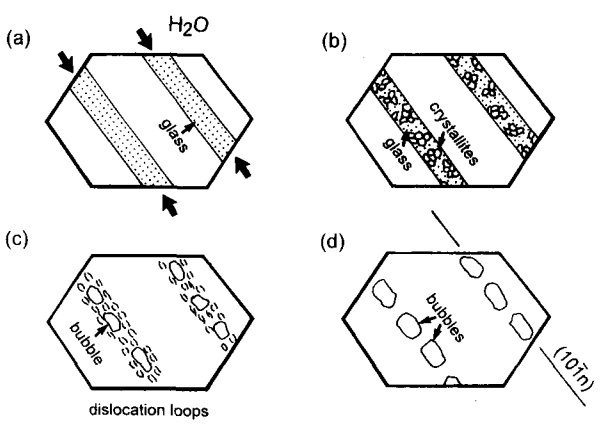


FIG. 8. Schematic representation of the development of decorated PDFs. (a) Close to initial state with amorphous (glass) PDFs containing dissolved H_2O. (b) Initial recrystallization to a mixture of glass and quartz crystallites. (c) Exsolution of H_2O from recrystallizing quartz in PDFs in the form of bubbles and adjacent dislocation loops. (d) Final state with original PDF now represented as a series of bubbles aligned along a specific crystallographic orientation. See text for more details.

At impact structures, diaplectic quartz glass and lechatelierite are also often recrystallized due to secondary annealing. There are two characteristic settings for these phases in which these thermal effects occur in lithic or mineral clasts: (1) incorporated into impact melt lithologies and (2) embedded in the central part of suevite breccia deposits. The recrystallization of the glass may be accompanied by the formation of cristobalite.

In the case of clasts in impact melt rocks, the most characteristic recrystallization texture of diaplectic glass is the "ballen texture" (Carstens, 1975; Grieve, 1975; Floran *et al.*, 1978). It was shown by Bischoff and Stöffler (1984) that the recrystallization texture depends on the pre-annealing shock stage of the clasts. Quartz with PDFs (<35 GPa) becomes brownish and dull and shows incipient formation of ballen texture. Diaplectic quartz glass from somewhat higher shock levels transforms to a texture in which the individual "ballen," consisting of α-quartz single crystals, all have the same optical orientation (Fig. 9). Diaplectic glass from higher shock levels shows ballen with different optical orientation, which gradually transform into ballen with intragranular polycrystallinity, if the original state of lechatelierite was reached (>50 GPa). The formation of the "ballen texture" may result from cristobalite, which forms in an intermediate stage of recrystallization before final transformation to α-quartz (Förstner, 1967; Carstens, 1975).

In suevite deposits, a large fraction of the glasses derived from quartz is commonly recrystallized (see papers in French and Short, 1968 and Bavarian Geological Survey, 1970, 1972 and 1977). For example, in the suevite breccia lens of the central crater cavity of the Ries structure, all glass is completely recrystallized due to hydrothermal activity at temperatures up to ~600 °C. Diaplectic glass and lechatelierite are only preserved in meter-thick zones at the bottom and top of the suevite deposits (10 m–25 m in thickness) outside of the rim of the impact structure (Engelhardt, 1972).

Annealing of quartz experimentally shocked to 20 GPa–30 GPa at 300 °C, 600 °C and 950 °C for 4 days indicates contraction of the originally expanded cell parameters, which are a result of the shock compression and release (Schneider and Hornemann, 1976). To our knowledge, no annealing experiments have been carried out on naturally shocked diaplectic quartz glass. In diaplectic quartz glass

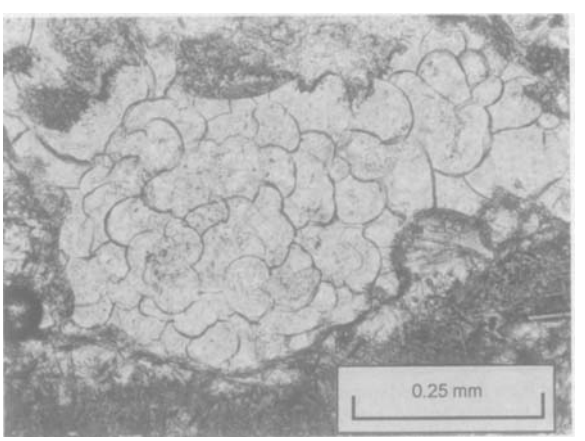


FIG. 9. Photomicrograph (plane polarized light) of ballen quartz in a clast within an impact melt rock from the Mien structure.

produced in shock recovery experiments, however, the crystal lattice is restored at temperatures > 1200 °C–1300 °C. The PDFs in quartz shocked to < 34 GPa persist up to this annealing temperature. The alteration of diaplectic glass depends on the degree of shock and the annealing temperature. The refractive index of the glass first increases at temperatures > 700 °C to 800 °C, before recrystallization starts at 1150 °C (Rehfeldt-Oskierski and Stöffler, 1986). The recrystallization texture is influenced by the shock pressure at which the glass was formed, as observed in nature (Bischoff and Stöffler, 1984). In any case, an assemblage of cristobalite and α-quartz is formed at > ~1200 °C in contrast to the annealing behavior of synthetic silica glass, which always transforms to only cristobalite. The recrystallization of diaplectic quartz glass starts at grain boundaries and forms a fibrous texture. Above 1200 °C, a ballen texture develops, similar to the one observed in natural impact formations; however, the ballen are always polycrystalline.

PRESSURE AND TEMPERATURE CALIBRATION

Shock Wave Barometry

As shock metamorphism results largely in disequilibrium because of the extremely short time scales, the classical concepts of barometry, such as mineral assemblages or equilibrium compositions of cogenetic minerals, cannot be applied. Three shock-induced properties (Stöffler, 1972, 1974) are useful for shock barometry: (1) the occurrence of pressure-specific shock features in minerals, (2) changes in physical (bulk) mineral properties and (3) the variation in the orientation of planar microstructures. The most recent (experimentally-derived) data base on shock barometry for quartz in crystalline rocks is comprehensively discussed in Stöffler and Langenhorst (1994) and is summarized in Fig. 1 and Table 3.

Shock barometry based on the occurrence of a specific shock effect is relatively imprecise since specific shock effects are often present over a pressure range. For example, planar microstructures in quartz occur at pressures between ~5 GPa and 35 GPa (Fig. 1; Table 3). The coexistence of two or more shock effects can, however, give a better pressure constraint than the observation of a single shock effect. For example, if quartz contains both PDFs and coesite, the pressure can be constrained to between ~30 GPa and 35 GPa (Fig. 1; Table 3). More precise shock pressure information is provided by changes in physical properties, such as refractive indices and density (Fig. 10); however, these changes occur only at > ~25 GPa and over a limited pressure interval. There are some shock recovery experiments calibrating changing x-ray properties in quartz (*e.g.*, Hörz and Quaide, 1973; Schneider *et al.*, 1984; Langenhorst, 1994). Other properties, such as the broadening of nuclear magnetic resonance, NMR, (Cygan *et al.*, 1990, 1992) show significant variations at <25 GPa but are not well calibrated by shock recovery experiments. When applying any of these calibration data to natural lithologies, there are two assumptions: (1) the time difference between natural and experimental shock events has no effect on the transformation of quartz and (2) the natural sample is unaffected by post-shock annealing or alteration. We have already discussed why the former is not a factor, except possibly in the synthesis of coesite and stishovite. The latter, however, is often not the case.

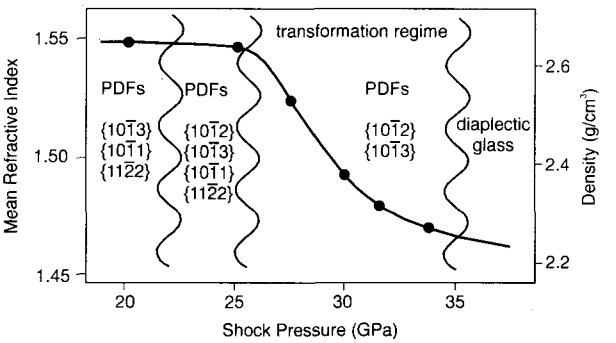


FIG. 10. Variation in mean refractive index and density of quartz shocked at >20 GPa at room temperature. Note the reduction in the values of these physical properties only occurs at > ~25 GPa in the regime where crystalline quartz is beginning to develop significant quantities of diaplectic glass. Also indicated are orientations of PDFs and their variation. (After Langenhorst and Deutsch, 1994.)

Fortunately, complications arising from post-shock history are not of concern, if the orientations of planar microstructures are used for pressure estimation. Even if PDFs have been annealed, their original orientation is still recognizable due to decorations (Fig. 3). In addition, if the current model for the formation of PDFs is correct (Goltrant *et al.*, 1992), the orientation of PDFs is determined within the shock front during initial loading. Consequently, any effects due to different pulse lengths in natural and experimental shock events should have no influence on the orientation. Thus, the orientation of planar microstructures is regarded as a reliable shock pressure indicator. Statistical data on the orientation of planar microstructures, as function of shock pressure, have been provided by several U-stage investigations of experimentally shocked quartz (Müller and Défourneaux, 1968; Hörz, 1968; Mitsjuk and Gorogotskaja, 1980; Gurov and Gurova, 1991; Huffman *et al.*, 1993; Langenhorst and Deutsch, 1994).

The most common procedure for estimating the average recorded shock pressure in a polycrystalline sample is to measure the orientation of planar microstructures with respect to the c-axis of quartz in random grains, usually 20 to 25, using the universal stage (*e.g.*, Robertson, 1975). If the sample contains quartz with no planar microstructures, these are also recorded. The method to determine the crystallographic orientation of planar microstructures is described in Stöffler and Langenhorst (1994) and is repeated briefly here. The poles to all sets of planar microstructures and the c-axis of quartz are plotted on a stereonet and, then, rotated to a c-axis vertical orientation. With all poles in this standard projection, a template containing the rational

TABLE 3. Progressive shock metamorphism and pressure-post shock temperature conditions of crystalline quartzo-feldspathic rocks, modified from Stöffler (1984).

Shock stage	Pressure range (GPa)	Post-shock temperature range (°C)*	Shock effects in quartz
0	<5/10	<100	Irregular fractures
Ia	5/10–20	100–170	PFs and PDFs
Ib	20–35	170–300	PDFs, reduced refractive index; stishovite and minor coesite
II	35–45	300–900	Diaplectic glass, coesite and traces of stishovite
III	45–60	900–1200	Diaplectic glass, coesite
IV	60–100	1200–2500	Whole rock melt
V	>100	>2500	Vapor

* at an ambient temperature of ~25 °C.
PFs = planar fractures; PDFs = planar deformation features.

crystallographic indices of known planar microstructure orientations (see Fig. 13, 14 and Table 3 in Stöffler and Langenhorst, 1994) is overlain and rotated about the c-axis to best fit the orientations. The template generally contains circles of uncertainty of ±5°, associated with each crystallographic orientation. The orientations of all sets of planar microstructures, most commonly PDFs, in each grain are recorded, and the grain, as a whole, is assigned an orientation corresponding to the highest pressure orientation present (*e.g.*, if both (0001) and {10$\bar{1}$3} PDF orientations are present (Table 1), the grain is listed as {10$\bar{1}$3}. With the orientations determined, an estimated shock recorded pressure is assigned, based on experimental calibration (*e.g.*, Table 4), to each grain and the recorded shock pressure for the sample is calculated by averaging the results from the recorded shock pressure estimates assigned to the individual quartz grains.

Although planar microstructure orientations are often displayed as histograms of angles, there is no convention to the binning interval, which is usually between 2° and 5° (*e.g.*, Fig. 11). This can lead to apparently different frequency distributions, depending on the binning interval. In addition, the vertical axis of such histograms is variously expressed in terms of absolute numbers of PDFs, within a particular angular interval, or as some relative number, such as frequency percent (Fig. 11). It should also be noted that, in the absence of data tables (*e.g.*, Engelhardt and Bertsch, 1969), the technique of solely recording the angle of the poles of planar microstructures to the c-axis and plotting histograms of angles, which has become popular in recent years (*e.g.*, Bohor *et al.*, 1987), does not uniquely yield the correct Miller indices. It should, hence, be discouraged. In addition, histograms express the raw angular information and usually take no account of observational error in determining the position of the c-axis, due to strain or even optical mosaicism, or the orientation of the planar microstructure. Reporting in tabular form or replotting the data after the orientations have been uniquely indexed using a template, which also allows for some accommodation of observational error, results in a much clearer rendition of the orientations observed (Fig. 12). This method of presentation, therefore, is recommended over histograms of angles (Fig. 11). The precision of measuring azimuthal and polar angles can be improved by use of a spindle stage because the position of the c-axis of quartz can be determined within ±1°. In addition, the spindle stage has the advantage over the U-stage in that a blind circle does not exist. The spindle stage technique, however, requires single grains instead of thin sections, and the recognition of the vertical position of a planar microstructure is more difficult because the spindle stage has only two axes of rotation.

A somewhat different scheme for estimating average recorded shock pressure using PDF orientations has been developed by Gurov and Gurova (1991). As with the Robertson (1975) scheme, the orientations of PDFs in a number of quartz grains are measured, and then the percentages of (0001), {10$\bar{1}$3} and {10$\bar{1}$2} orientations are calculated for the sample. They assign 5 GPa for the development of (0001), 10 GPa–13 GPa for {10$\bar{1}$3} and 15 GPa–18 GPa for {10$\bar{1}$2} features (*cf.*, Table 4) to calculate the average recorded shock pressure. Apart from the assigned pressures for the various orientations, the main difference in this technique is that no account is taken of quartz grains in the samples that show no PDFs. In the Robertson (1975) scheme, they are assigned a value of 3.5 GPa, provided the samples under examination occur within the outer limit of shatter cone formation. Fel'dman (1994) extended the Robertson (1975) scheme to estimate the recorded shock pressures in quartz from Janisjärvi and Popigai and in the 5 km core from the central uplift of Puchezh-Katunki in order to provide a calibration for various shock metamorphic effects in mafic minerals. He used the pressure calibration in Grieve and Robertson (1976) but also considered the number of quartz grains converted to diaplectic quartz glass to which he assigned a recorded shock pressure of 30 GPa (*cf.*, Fig. 1).

TABLE 4. Shock pressures required for onset of various shock effects in quartz for moderate pre-shock temperatures (<250 °C) and in non-porous lithologies, after Stöffler and Langenhorst (1994).

Shock effect	Pressure (GPa)
Planar fractures: (0001), {10$\bar{1}$1}	>5
Planar deformation features: (0001)	>5
Planar deformation features: {10$\bar{1}$3}	>10
Planar deformation features: {10$\bar{1}$2}	>20
Stishovite	>12–15
Coesite	>30
n_{quartz} = 1.52–1.47	28–34
Diaplectic glass (n = 1.468–1.460)	35–50
Lechatelierite	>50

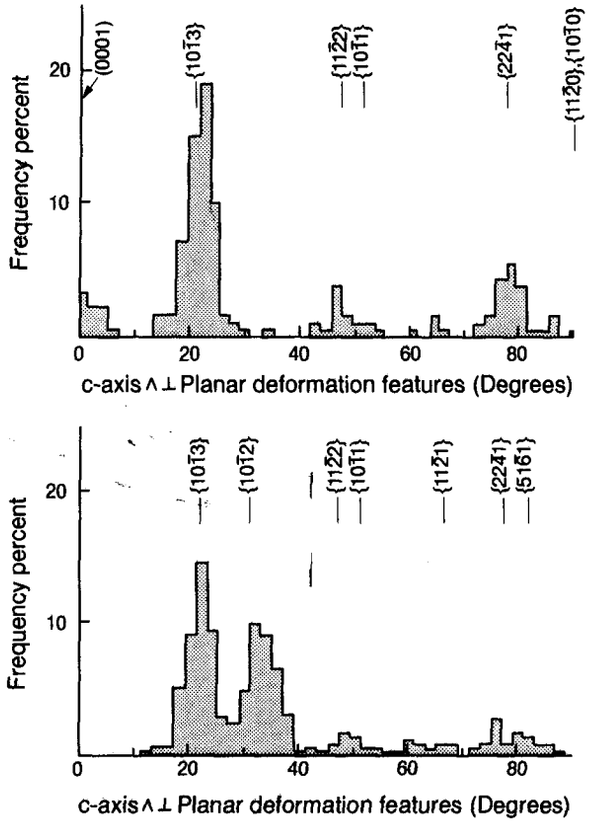


FIG. 11. Histograms of frequency percent of angles, binned at 2°, between the c-axis and poles to the planes of decorated PDFs from (top) so-called type C shock deformation in quartz from seven Canadian impact structures. Note the occurrence of both (0001) and {10$\bar{1}$3} orientations and (bottom) so-called type D shock deformation in quartz from 10 Canadian impact structures. Note the occurrence of both {10$\bar{1}$3} and {10$\bar{1}$2} but absence of (0001) orientations. Data are from Robertson *et al.* (1968).

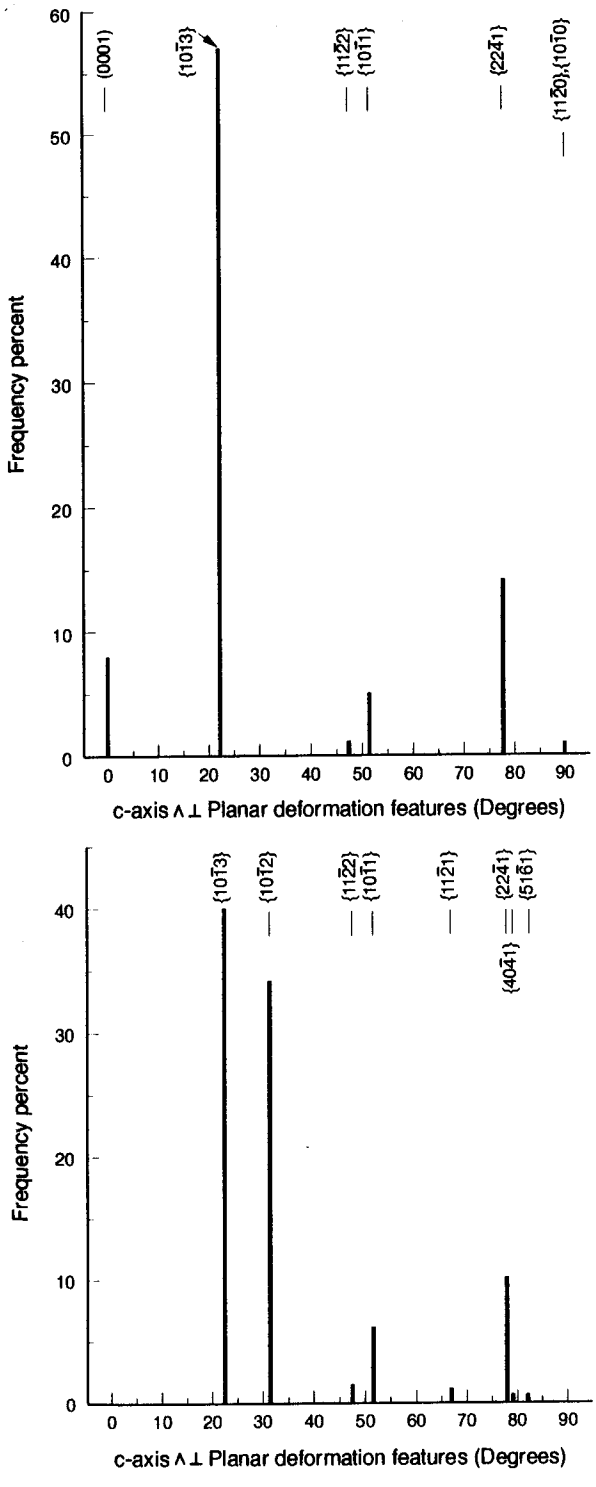


FIG. 12. As in Fig. 11, except PDFs have been uniquely assigned specific Miller indices through the use of a template containing known planar microstructure orientations. In type C and type D deformations, 14% and 7% of the orientations, respectively, could not be indexed uniquely. This represents angular data on PDF orientations with respect to the c-axis with observational error exceeding the ±5° uncertainty circles associated with the template (see Fig. 13 in Stöffler and Langenhorst, 1994). Compare with Fig. 11.

The refractive indices of quartz with PDFs and of diaplectic glass provide the most accurate pressure calibration for quartz shocked above ~25 GPa (Fig. 10 in this work and Fig. 29a,b in Stöffler and Langenhorst, 1994). An analogous correlation holds for the density of shocked quartz (Fig. 10; Langenhorst, 1994). In principle, the calibration scale for non-porous, quartz-bearing rocks can be extended to ~50 GPa on the basis of refractive index data (Fig. 1; Table 1 and 4), most usefully by monitoring the texture and refractive index of coexisting feldspar glasses (Stöffler, 1971b, 1974; Lambert, 1981; Lambert and Grieve, 1984). In these cases, however, it is important to exclude any post-impact thermal effects on physical properties.

The pressure calibration data for planar microstructure orientations were mainly obtained by shock experiments on single crystals (*e.g.*, Hörz, 1968; Müller and Défourneaux, 1968; Langenhorst and Deutsch, 1994). Although some experiments have been undertaken, there are no systematic experimental data for polycrystalline rocks. Thus, although the relative order of appearance of planar microstructure orientations in nature and experiments may be equivalent, there is no guarantee that the absolute pressure estimates for natural events are equivalent. Furthermore, the influence of porosity on the development of planar microstructures has never been investigated experimentally. The effect of porosity, particularly on the relative abundance and orientations of planar microstructures, has been documented in a number of natural cases (*e.g.*, Bunch and Cohen, 1964; Stöffler, 1972; Robertson, 1980; Grieve and Therriault, 1995). It is most obvious in various types of sandstones and is discussed in some detail later.

Shock Wave Thermometry

There are three different temperatures in the context of shock metamorphism: (1) the pre-shock, (2) the shock and (3) the post-shock temperature. The latter two are correlated with shock pressure. The porosity of the target rocks has considerable influence, as shock temperatures in porous rocks can be an order of magnitude higher than in compact crystalline rocks (Ahrens and Gregson, 1964; see also Figs. 1, 2, 6 in Stöffler and Langenhorst, 1994). For non-porous targets, the shock and post-shock temperatures of quartz are known through various calculations (Wackerle, 1962; Mashimo *et al.*, 1980; Langenhorst, 1994) and a few pyrospectrometric experimental measurements (Raikes and Ahrens, 1979; Lyzenga and Ahrens, 1980; Sugiura *et al.*, 1982; Lyzenga *et al.*, 1983). Thus, the shock and post-shock temperatures in quartz can be derived if the pressure has been previously estimated on the basis of shock effects. A direct estimate of these temperatures is not possible since quartz does not exhibit any shock effect that is dependent on temperature and independent of pressure. However, with shock effects in quartz providing the pressure estimate, thermal effects in other (*e.g.*, mafic) minerals can be used to estimate temperatures (*e.g.*, Fel'dman, 1994). In addition, melting phenomena in quartz and coexisting vesiculated feldspar glasses indicate that, at a minimum, the melting temperatures of these phases have been reached, at corresponding peak pressures of >50 GPa and 45 GPa, respectively.

Recently, experimental studies have focussed on the influence of the pre-shock temperature on shock effects in quartz (Reimold, 1988; Gratz *et al.*, 1992a; Langenhorst *et al.*, 1992; Huffman *et al.*, 1993; Langenhorst, 1994; Langenhorst and Deutsch, 1994). These studies were stimulated, to some degree, by the debate over the

Vredefort structure, where the superposition of shock and thermal metamorphism has taken place. It has been suggested that higher pre-shock temperatures could significantly lower the threshold pressure of PDF formation (Carter *et al.*, 1990; Officer, 1992), although experimental data indicate that this is the diagnostic shock effect least affected by temperature (Huffman *et al.*, 1993). Despite this interest in the pre-shock temperature, only one crystallographic criterion based on the analysis of PDF orientations has turned out to be useful for thermometry. Quartz shocked at pre-shock temperatures in the α-stability field (<573 °C) exhibits PDF orientations that belong only to one rhombohedron (*e.g.*, $\{10\bar{1}3\}$); that is, corresponding rhombohedra (*e.g.*, $\{10\bar{1}3\}$ and $\{01\bar{1}3\}$) never occur together (Langenhorst and Deutsch, 1994). In contrast, shock at pre-shock temperatures in the β-stability field (>573 °C) can produce PDFs parallel to planes of hexagonal pyramids. With this crystallographic criterion, it may be possible to distinguish whether quartz was shocked at pre-shock temperatures in the α- or in the β-stability field. A detailed analysis of PDF orientations in quartz from the Ries structure (Engelhardt and Bertsch, 1969) indicates PDF planes belonging only to rhombohedral forms (*i.e.*, PDF poles of one form define 120° angles in a stereoplot). Pre-shock temperatures in Ries target rocks were, thus, <573 °C, as expected from the size of the structure. Only in cases of very large-scale impact would rocks in the deeper crust, at temperatures >573 °C, be affected by shock (Langenhorst *et al.*, 1992).

NATURAL OCCURRENCES OF SHOCKED QUARTZ

"Shocked" quartz occurs in the full range of impact lithologies at terrestrial impact structures, including parautochthonous target rocks of the crater floor, in clasts in allochthonous lithologies, such as breccias and impact melt rocks, and in various types of proximal and distal ejecta (Fig. 13). In parautochthonous rocks, the distribution of shocked quartz can be used to estimate rates of shock pressure attenuation (*e.g.*, Robertson and Grieve, 1977) and derive constraints on cratering mechanical models (Ivanov, 1994). In allochthonous lithologies, the utility of the identification of shocked quartz is mainly to confirm an impact origin for a particular lithology (*e.g.*, Bohor *et al.*, 1984; Gostin *et al.*, 1986) and/or structure (*e.g.*, Engelhardt *et al.*, 1967; Halls and Grieve, 1976; Lambert *et al.*, 1980, 1981).

Impact Structures in Crystalline (Non-porous) Targets

Planar Microstructures–There have been surprisingly few systematic and detailed studies of planar microstructures and the variation in the recorded level of shock in quartz in impact structures formed in crystalline targets. Planar fractures are seldom described in detail, and virtually all studies have used observations of the orientation of PDFs as a guide to variations in recorded shock pressure. Individual quartz grains often contain a variety of PDF sets of different orientation, with the number of sets increasing with recorded shock pressure and, within a single sample, individual quartz grains can record different levels of shock. For example, grains with the (0001) orientation can coexist with grains with $\{10\bar{1}3\}$, which in turn can co-exist with $\{10\bar{1}2\}$ orientations (*e.g.*, Table 5; Robertson *et al.*, 1968; Engelhardt and Bertsch, 1969). Basal PDFs (0001), however, do not apparently exist in the same grain as $\{10\bar{1}2\}$ PDFs (Table 5). Individual quartz grains with (0001), however, can coexist with grains with $\{10\bar{1}2\}$ features in the same sample, but this is relatively rare (Robertson, 1975). As a result of this variation in PDF orientations in individual grains, average recorded shock pressure estimates for a sample, derived from the orientations in individual grains, have a range of ~25% (Robertson and Grieve, 1977). The variation between individual quartz grains is due to irregularities in the shock wave that are a result of the polycrystalline nature of rocks, which causes local stress variations. From experimental results on single crystals of quartz, it is also known that the orientation of the individual quartz grains with respect to the shock wave has some effect on PDF orientations (Hörz, 1968; Langenhorst and Deutsch, 1994).

Some of these variations are well illustrated in samples from the central uplift of the Slate Islands structure, which has a rapidly varying geology of felsic and mafic Archean metavolcanics and intrusives, Proterozoic metasediments and metavolcanics and diabase dikes (Sage, 1991). Here, there are small differences (~2 GPa) in the estimated average recorded shock pressure derived from PDF orientations between clusters of quartz grains of differing grain size in the same sample. There are also differences in recorded shock levels between isolated and clusters of grains in the same sample, as well as variations between adjacent samples of felsic and mafic rock types (Grieve and Robertson, 1976; Robertson and Grieve, 1977). Similar variations, with larger quartz grains having more sets of PDFs and, overall, more PDFs have been noted at other impact structures (*e.g.*, Walzebuck and Engelhardt, 1979; Sazonova, 1981). In general, these variations have not been systematically studied and are likely due to reverberations and impedance differences and variations in orientation and shock pulse length within individual grains. They are also not sufficient, however, to mask general trends in the overall spatial variation of PDF development at impact structures (*e.g.*, Robertson and Grieve, 1977).

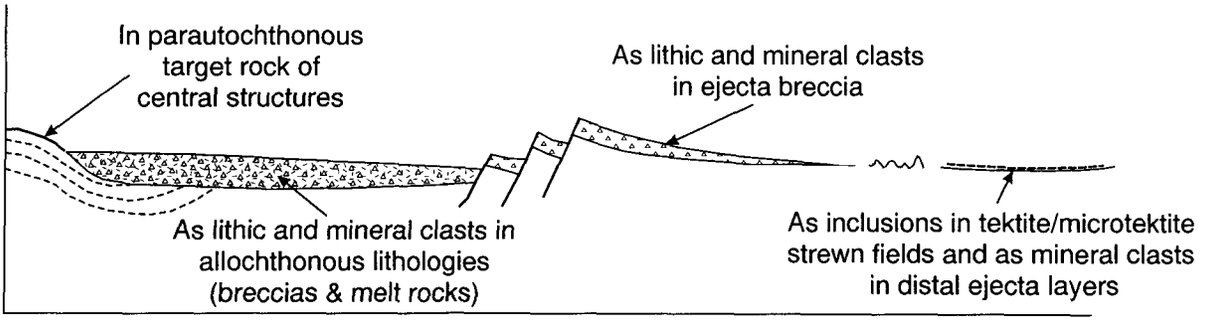


FIG. 13. Schematic half-space cross-section of the potential spatial distribution of quartz showing shock metamorphic effects at a terrestrial complex impact structure. Schematic is not to scale, particularly with regard to horizontal dimensions. Dashed lines in parautochthonous target rocks of central uplift indicate distribution of shock isobars.

TABLE 5. The PDF orientations and average recorded shock pressures in parautochthonous rocks of the crater floor at Brent, estimated from PDF orientations, in drill hole 1-59 (after Robertson and Grieve, 1977).

Depth[1] (m)	% quartz with[2] planar deformation features	% grains[3] with (0001) orientation	% grains[3] with {10$\bar{1}$3} orientation	%grains with[3] {10$\bar{1}$3}, + {10$\bar{1}$1} {11$\bar{2}$2} or {22$\bar{4}$1} orientations	% grains[3] with {10$\bar{1}$2} orientation	Average[4] pressure (GPa)
1103	100	–	–	–	100	23.0
1106	100	–	5	5	90	22.1
1107	100	–	–	15	85	21.8
1108	100	–	5	10	85	21.7
1109	100	–	10	10	80	21.1
1110	100	–	20	35	45	18.0
1112	100	–	35	15	50	18.0
1113	88	–	12	38	38	16.5
1114	95	–	85	5	5	12.4
1116	100	–	95	5	–	12.2
1117	95	–	75	10	–	13.1
1122	95	–	90	5	–	11.8
1123	95	–	70	25	–	12.4
1130	85	25	60	–	–	10.2
1137	50	35	10	5	–	7.8
1140	50	45	5	–	–	7.3
1142	58	58	–	–	–	7.4
1147	55	10	45	–	–	8.8
1159	40	35	5	–	–	7.0
1161	31	23	8	–	–	6.8
1169	40	35	5	–	–	7.0
1176	30	30	–	–	–	6.5
1182	15	15	–	–	–	6.0
1188	5	5	–	–	–	5.7

[1] Depths are from estimated impact point, 220 m above present ground surface.
[2] Orientations measured in 20 grains per sample.
[3] PDF orientation recorded represents highest pressure feature in each grain. Other orientations may be present (*e.g.*, grain recorded as {10$\bar{1}$3} may also have (0001) orientation). Note the subdivision of PDF orientations is similar to Table 1.
[4] See Robertson and Grieve (1977) for details and calculation of average recorded shock pressure.

To our knowledge, there has been only one case of the systematic study of variations in PDF development in quartz in the parautochthonous rocks beneath a simple impact crater in a crystalline target and this is at the Brent structure (Dence, 1968; Dence *et al.*, 1977; Robertson and Grieve, 1977). Shock effects occur in the parautochthonous rocks of the floor of the impact structure only in the center (Fig. 14). For ~15 m beneath the impact melt lens at the base of the allochthonous breccia lens, all indications of shock metamorphism in quartz have been recrystallized. Below this depth, however, {10$\bar{1}$2} PDFs dominate but progressively decrease in abundance with depth, being replaced by {10$\bar{1}$3}, then (0001) as the dominant orientation. Finally, at a depth of 85 m below the floor of the impact structure, no PDFs are observed (Table 5; Dence, 1968). Estimation of the shock pressures recorded by the PDFs ranges from 23.0 GPa to 5.7 GPa over the 85 m (Table 5; Fig. 15; Robertson and Grieve, 1977). The rate of shock pressure decay is $\sim R^{-40}$ at higher recorded shock levels, reducing to $\sim R^{-7}$ at lower recorded shock levels, where R is radial distance normalized to the transient cavity radius (Robertson and Grieve, 1977). These extremely high rates (Fig.15) do not take into account that the entire section of rock is parauthochthonous and has been sheared out and thinned as it was driven down in the formation of the transient cavity. The upper portions were displaced downward by ~500 m and thinned by 50% during cavity formation (Robertson and Grieve, 1977). This is in concert with "before" and "after" observations at underground nuclear explosions in crystalline targets (Borg, 1972),

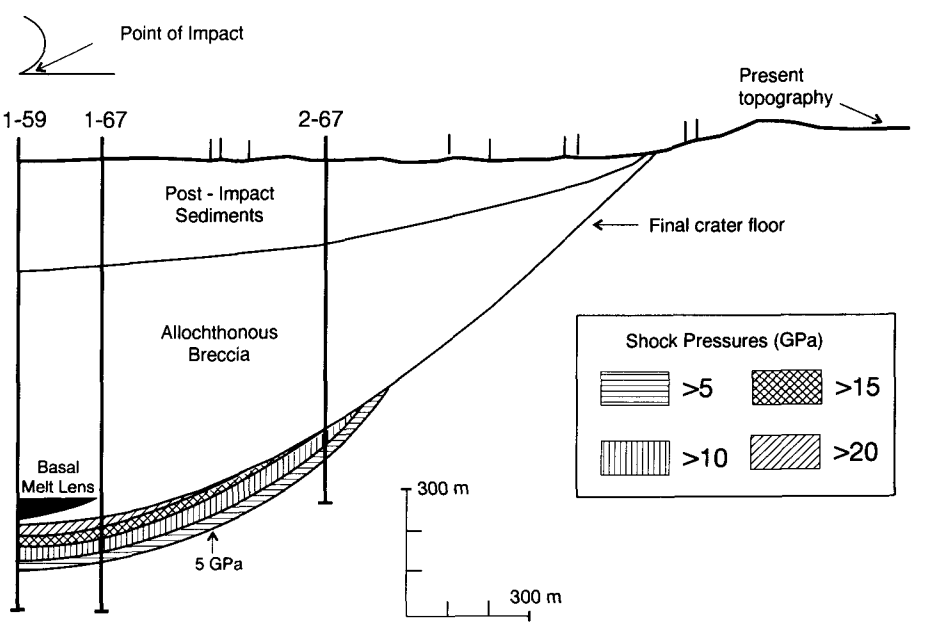


FIG. 14. Schematic half-space cross-section of the Brent impact structure, showing location of drill holes, which penetrated impact lithologies into the parautochthonous target rocks. Note the spatial restriction of the occurrence of shocked parautochthonous target rocks to the central area of true floor of the impact structure. Also indicated are the zones recording various shock pressures on the basis of the orientation of PDFs in quartz. Shocked quartz also occurs in mineral and lithic clasts in the allochthonous breccia partially filling the structure, although most of the breccia is not obviously unshocked. See Grieve (1978) and text for details.

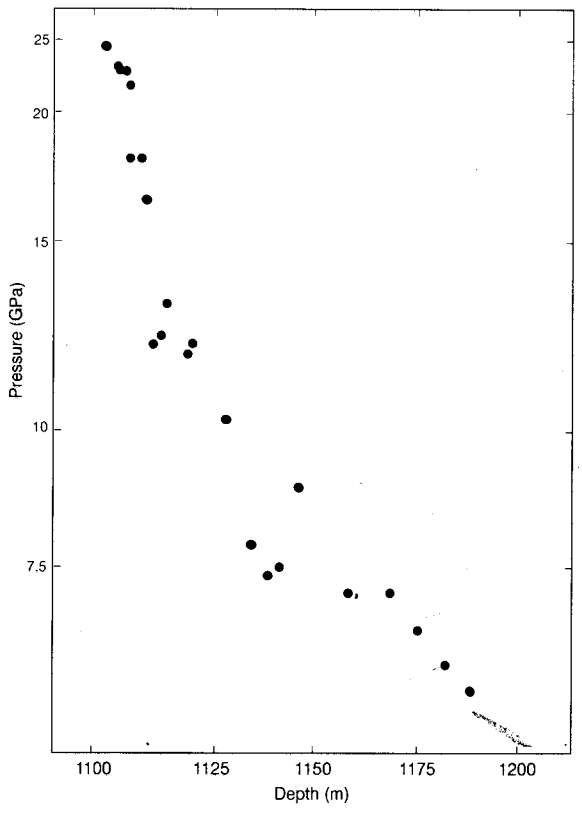


FIG. 15. Variation in recorded pressure in quartz in parautochthonous target rocks, based on PDF orientations, in drill hole B1-59 at the Brent structure (see Fig. 14 for location of drill hole). Note that the vertical scale is logarithmic, see text for details. See also Table 5. (After Robertson and Grieve, 1977.)

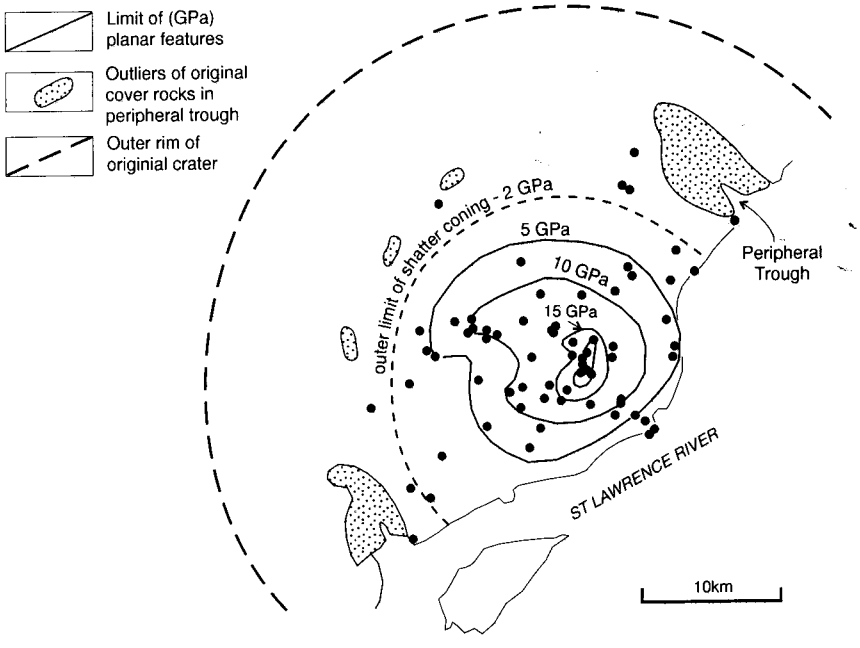


FIG. 16. Distribution of samples used to derive surface shock pressure contours of PDFs in quartz at the Charlevoix structure. Outer limit of PDF development is within that of shatter coning and well inside that of the original rim. Also indicated are locations of various recorded shock isobars, which were estimated largely from PDF orientations. (After Robertson and Grieve, 1977.)

the result of cratering experiments into multi-layered colored sands (Stöffler *et al.*, 1975), and recent finite difference models (O'Keefe and Ahrens, 1993). When these displacements are taken into account, the calculated attenuation rates are of the order of R^{-2} to R^{-3} (Robertson and Grieve, 1977), similar to axial attenuation rates calculated in finite difference models (Ahrens and O'Keefe, 1977).

Short (1970a) used PDFs in quartz as a parameter in creating a "shock index" for the allochthonous breccia lens at the West Hawk structure. From the shock index, he noted a variation in the level of shock metamorphism with maximum shock levels occurring towards the top and base of the breccia lens. A similar situation occurs in the breccia lens at Brent (Dence *et al.*, 1977; Grieve, 1978). Most of the quartz, however, in the breccia lenses at simple impact structures does not display obvious shock metamorphic features, and the range of shock levels is consistent with the allochthonous nature of the breccias. The concentrations of more highly shocked materials towards the base and upper zones in the center of simple impact structures are due to movements associated with the inward collapse of the transient cavity wall and its lining of more highly shocked materials during transient cavity modification (Dence *et al.*, 1977; Grieve *et al.*, 1977; Melosh, 1989). To our knowledge, there have been no studies of the shock metamorphism of quartz in the fallback layer that lies on top of the allochthonous breccia lens at simple structures in crystalline targets.

At complex impact structures, such as Charlevoix and the Slate Islands (Robertson and Grieve, 1977), variations in the orientations of PDFs in quartz indicate that recorded shock levels in the exposed parautochthonous rocks of the crater floor attenuate from ~25 GPa at the center to <5 GPa at radial distances of <<0.5 of the rim diameter (Fig. 16). These are minimum estimates as these structures have been eroded below the original crater floor, but they do emphasize the observation that shock metamorphic effects in quartz in parautochthonous target rocks of complex structures are, like those at simple structures, spatially limited to the central area (Fig. 16). The calculation of an attenuation rate from these observations is complicated by the fact that complex structures are highly modified crater forms, with respect to the original geometries of their transient cavities (*e.g.*, Dence *et al.*, 1977; Croft, 1980, 1981; Grieve *et al.*, 1981; Melosh 1989). Nevertheless, geometrically constrained reconstructions of the transient cavity at complex structures indicate that the development of shock features in quartz is not only restricted to well within the original rim (Fig. 16) but also to within the transient cavity (Fig.17) and give reasonable attenuation rates that are similar to those for simple structures (Robertson and Grieve, 1977). These types of relationships permit the relative spatial limitation of quartz with PDFs to be used to estimate original morphologic parameters at heavily eroded structures, where there is little or no indication of the original rim. This has been done with respect to both transient cavity and final rim diameters (Fig. 18; Stöffler *et al.*, 1988; Lakomy, 1990) and, most recently, at very large, eroded complex structures, such as Sudbury (Grieve *et al.*, 1991; Deutsch and Grieve, 1994; Stöffler *et al.*, 1994) and Vredefort (Therriault *et al.*, 1993).

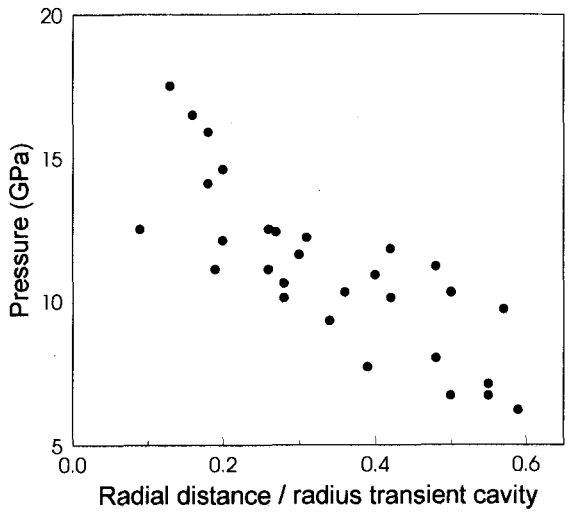


FIG. 17. Variation in recorded shock pressure, as estimated from PDF orientations in quartz at Charlevoix, relative to radial proportion, which is defined as radial distance from the center normalized to transient cavity radius. Note that PDF development is restricted to within the transient cavity. This is similar to the case at simple structures (Fig. 14) where there has been considerably less modification of the transient cavity. (After Robertson and Grieve, 1977.)

Vredefort represents a special case, with respect to the spatial distribution of PDFs of various orientations in the parautochthonous rocks of the crater floor. Unlike other impact structures, there is no well-defined progressive change from (0001) to $\{10\bar{1}3\}$ to $\{10\bar{1}2\}$ orientations being relatively more abundant as the center of the structure is approached. By far the dominant observation is the occurrence of decorated (0001) PDFs (Carter, 1965, 1968; Fricke *et al.*, 1990; Grieve *et al.*, 1990). This anomalous appearance and spatial distribution of PDFs at Vredefort has been used to question its impact origin and even the occurrence of *bona fide* PDFs (Reimold, 1987, 1990; Reimold and Wallmach, 1991).

It has been proposed that the anomalous distribution of PDFs is due to the effects of recrystallization, which generally increases

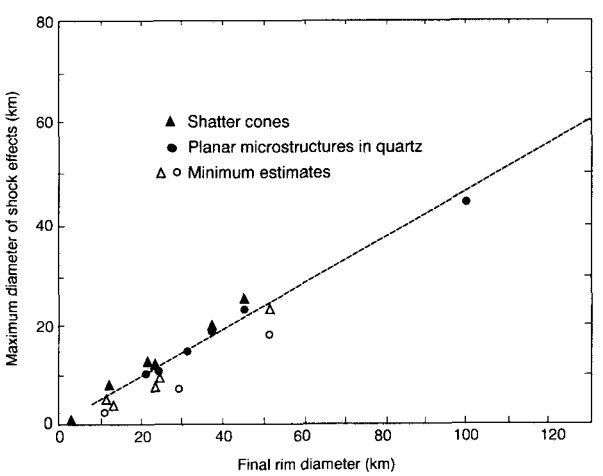


FIG. 18. Maximum diameter of occurrence of shock effects (shatter cones, planar microstructures in quartz) relative to final rim diameter at complex terrestrial impact structures. Such a diagram can be used to estimate final rim diameters, given data on the maximum spatial distribution of planar microstructures in quartz, as approximated by dashed line. Note that the shatter cones invariably occur out to greater diameters than planar microstructures in quartz. (After Lakomy, 1990.)

towards the core of the deeply eroded Vredefort structure (Grieve *et al.*, 1990; Hart *et al.*, 1991; Martini, 1992). The effect of the recrystallization has been to anneal out, in many cases, virtually all evidence of PDF development, including decorations. In some cases, however, a few remanent $\{10\bar{1}3\}$ orientations remain and may even persist across the grain boundaries of the newly formed quartz (Fricke *et al.*, 1990). Orientations parallel to $\{10\bar{1}2\}$ are very rare (Grieve *et al.*, 1990). Due to the recrystallization, the Robertson (1975) scheme (or some variant of it) for estimating recorded shock pressure can not be used, as the potentially higher recorded shock pressures correlate with the higher degrees of recrystallization towards the center of the structure. If, however, the PDF orientation corresponding to the highest recorded shock pressure in any grain is taken to represent the average recorded shock pressure for the entire sample, then there is a poorly defined but progressive increase in recorded shock pressure as the center of Vredefort is approached (Grieve *et al.*, 1990). Recent TEM studies have confirmed the occurrence of basal (0001) PDFs at Vredefort (Leroux *et al.*, 1994). They also observed the effects of thermal annealing and noted that the basal PDFs are more robust to the effects of annealing compared to other orientations, because they are twins. They also recorded the occurrence of coesite and stishovite associated with thin veins of pseudotachylite (White, 1993; Leroux *et al.*, 1994).

Systematic studies of PDF orientations in quartz with depth at complex structures are limited. This is generally due to sampling problems, since few drill cores of sufficient depth are available. Studies at the Ries structure indicate that the intensity of PDF development and the recorded shock pressure attenuate over ~600 m in the crystalline basement of the crater floor (Engelhardt and Graup, 1977). This study also indicates structural complexities in the floor of the impact structure with depth and that, as a whole, the drilled section consists of parautochthonous materials within which there has been faulting and displacement. Similarly, the recorded shock pressure, as determined from PDF orientations, decreases with depth in the parautochthonous rocks of the uplifted floor of the Kara structure (Sazonova, 1981; Sazonova *et al.*, 1981; Gurov and Gurova, 1991). By far, the most spatially extensive study is at the Puchezh-Katunki structure, where samples from 5 km of core drilled in the center are available (Pevzner *et al.*, 1992). The estimated recorded shock pressure levels decrease from ~40 GPa at the top of the central uplift to ~10 GPa at 5 km (Fig. 19) and are largely based on variations in PDF orientations in quartz (Ivanov, 1994). As with the variations in recorded shock pressures in surface exposures of the crater floor, these values cannot be used directly to calculate an attenuation rate. The material that constitutes the central uplift was first driven downwards during transient cavity formation and then uplifted during cavity modification. Attempts to simulate this with a two-dimensional numerical code result in calculated recorded shock pressure gradients generally similar to the observed gradients (Fig. 19; Ivanov, 1994).

Studies on the distribution of shocked quartz within the ejecta blanket of complex craters are restricted to the Ries structure. In general, a decrease of the recorded shock level in quartz from top to bottom has been observed (Pohl *et al.*, 1977; Schneider, 1971). If suevite is present on top, it contains abundant shocked quartz ranging from lechatelierite to quartz with PFs of the lowest shock stage. This assemblage changes discontinuously at the base of suevite, where a polymict breccia (called Bunte breccia at the Ries) forms the main mass of the ejecta blanket. Schneider (1971) found

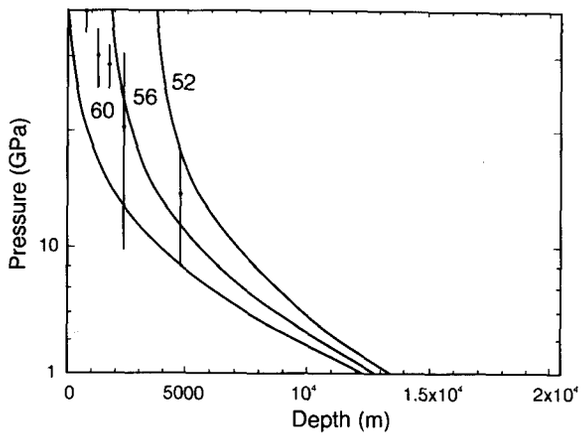


FIG. 19. Variation of recorded shock pressure (small dots), based on PDF orientations, with depth at the Puchezh-Katunki structure. Indication of range of pressures recorded in each sample (vertical bars). Lines represent time-evolution profiles of pressure during the modification stage of crater formation, in seconds after impact, based on two-dimensional-code calculations. Note the similarity of the trend of observed data points *vs.* calculated curves. (After Ivanov, 1994, and see same for more details of the significance of these data for cratering mechanical models.)

that the maximum shock level below the suevite is represented by diaplectic quartz glass and the abundance of quartz with PFs and PDFs continuously decreases with increasing depth in the Bunte breccia. Subsequent work has confirmed the occurrence of diaplectic quartz glass but found no systematic trend in the variation of recorded shock with vertical position (Banholzer and Hörz, 1979).

Diaplectic Quartz Glass, Lechatelierite, Coesite and Stishovite–Diaplectic glass, lechatelierite, coesite and stishovite occur at terrestrial impact structures in crystalline targets. Diaplectic glass is volumetrically the most common. It is, however, much less abundant than quartz with planar microstructures. Diaplectic glass requires higher shock pressures for its formation than planar microstructures (Fig. 1; Tables 1, 3, 4) and, thus, occurs over a smaller volume of the target rocks. In addition, diaplectic glass is a metastable phase, which can revert with temperature and time.

Diaplectic quartz glass occurs in suevite breccias and, less frequently, other polymict breccias (*e.g.*, papers in French and Short, 1968) but is rare in the parautochthonous rocks of the floors of terrestrial impact structures. It is unknown from the floors of simple structures and in complex structures has been reported only at the larger examples (*e.g.*, Manicouagan, Dressler, 1990; Puchezh-Katunki, Masaitis *et al.*, 1980, and Fel'dman, 1994). It is only at the largest structures that shock pressures are of sufficient magnitude in the parautochthonous rocks of the crater floor to result in diaplectic quartz glass (Figs. 1, 10 and Tables 3, 4). The increase in recorded shock pressure in the rocks of the floors of complex impact structures, with increasing diameter, is the result of the differential scaling of crater dimensions with respect to shock pressure attenuation, due to decreasing cratering efficiency at larger diameters (Melosh, 1989; Grieve and Cintala, 1992).

Lechatelierite is not generally observed in parautochthonous rocks at impact structures formed in crystalline targets. The pressures required for its formation (>50 GPa; Tables 3, 4) are in the same range as those required for whole-rock melting of crystalline rocks (Ahrens and O'Keefe, 1977; Orphal *et al.*, 1980; Grieve and Cintala, 1992). Lechatelierite may occur, however, as schlieren in impact melt rocks consisting of mixed glasses or in clasts in suevite. In some cases, where the impact melt rocks are particularly high in SiO_2, lechatelierite can occur as chemical "ghosts" in the glassy impact melt rocks. For example, at Wanapitei, the composition of matrix of the glassy melt rocks averages ~75% SiO_2 but includes areas of 100% SiO_2, after lechatelierite, that are not resolvable optically (Grieve and Ber, 1994).

Coesite and stishovite are generally recovered from relatively highly shocked clasts from breccias, which have cooled quickly. The occurrence of coesite has been reported from ~15 terrestrial impact craters in crystalline or mixed crystalline-sedimentary targets. Stishovite has also been described at approximately half as many structures. Coesite can be detected optically, in some cases, as highly refractive aggregates up to 100 μm–200 μm, generally in diaplectic quartz glass or, less commonly, in quartz with lower birefringence and multiple sets of PDFs. The grain size of the individual coesite crystals, which are typically twinned parallel to (010) (Fig. 5), is of the order of 300 nm. Stishovite is very seldom detected optically and appears most often as concentrations in the lamellae of PDFs (Stöffler, 1971a). Coesite is always more abundant than stishovite (Stöffler, 1971a; Martini, 1991). Coesite and stishovite have never been identified in parautochthonous lithologies at terrestrial impact structures, with the exception of Vredefort. Presumably, this is due to their metastable nature, with stishovite being less stable than coesite, and the relatively high shock pressures required for the formation of coesite (Tables 3 and 4). For example, at Ries, stishovite does not occur in suevite breccias where the PDFs are decorated (*i.e.*, where there is evidence of annealing; Stöffler, 1971a). Reversion of stishovite occurs experimentally in as little as 3 days at 250 °C in the presence of water (Gigl and Dachille, 1968).

Coesite and stishovite have generally been identified at relatively young impact structures. The exceptions to this are reported occurrences at a number of relatively old structures, such as Janisjärvi (Fel'dman *et al.*, 1978, 1979) and Vredefort (Martini, 1978, 1991). In the case of the Janisjärvi structure, the identification of coesite is based on the occurrence of only three x-ray lines and that of stishovite on two lines, and neither phase was detected optically (Fel'dman *et al.*, 1979). As coesite has been misidentified previously (*e.g.*, at the non-impact Richat structure by Cailleux *et al.*, 1964) by limited numbers of x-ray lines, it is possible that this is also the case at Janisjärvi.

Vredefort represents an unusual case. Coesite and stishovite occur in association with thin pseudotachylite veins, concentrated in metasediments in the northern sector of the collar rocks (Fig. 20). Although, it was originally thought that they occur only in clasts in the pseudotachylite (Martini, 1978), it is now apparent that they occur also in the quartzite wall rock at distances of <2 mm from the contact with the pseudotachylite (Martini, 1991). The presence of coesite has been confirmed in recent TEM studies (White, 1993; Leroux *et al.*, 1994). Stishovite was confirmed in the study by White (1993) and has also been detected by NMR (McHone and Nieman, 1988). Coesite occurs optically in what are, apparently, single crystals up to 150 μm across replacing quartz and as needles up to 100 μm long. Stishovite can also occur in crystals with similar habit and dimensions and be resolved optically (Martini, 1991). The relatively large grain size of coesite and stishovite at Vredefort can probably be attributed to the magnitude of the event, relative to most other known impact structures and, thus, relatively long times of both shock compression (formation of stishovite) and rarefaction (formation of coesite) (Stöffler, 1971a; Stöffler and

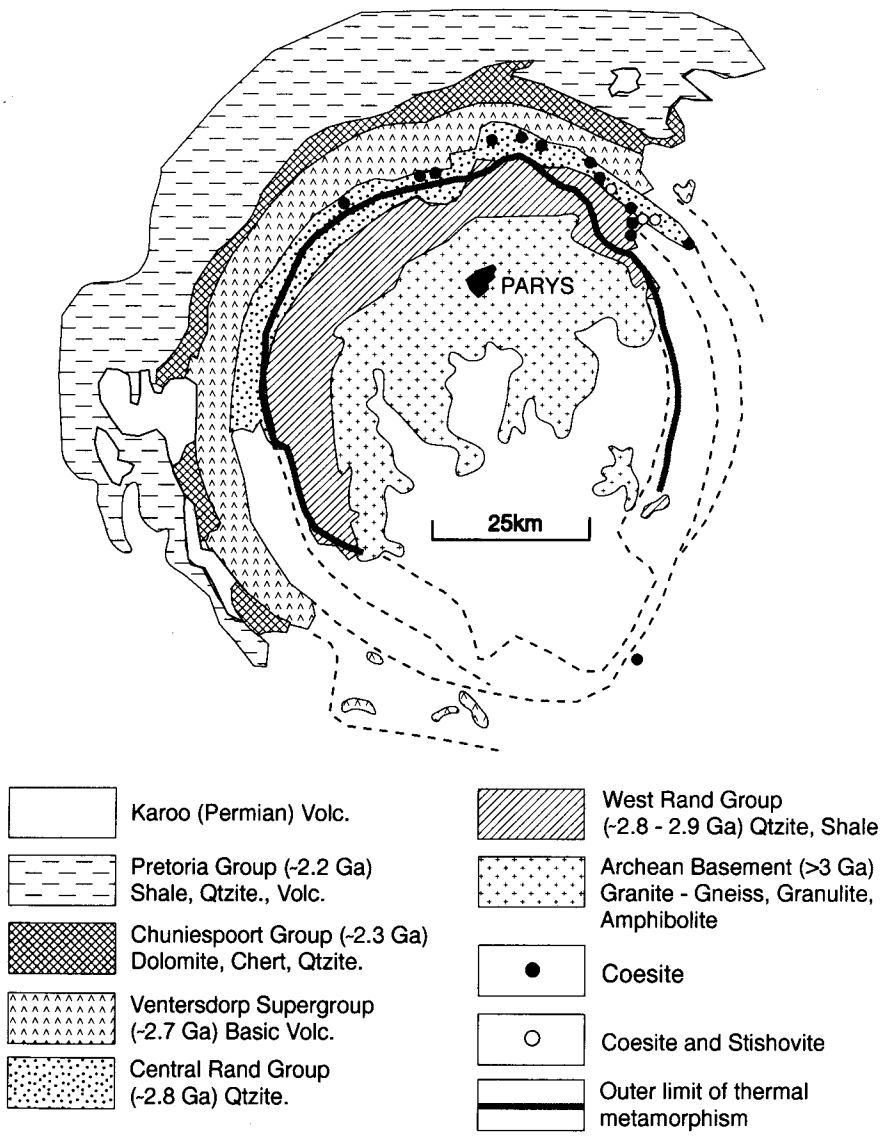


FIG. 20. Geologic sketch map of the Vredefort Dome (the center of the Vredefort impact structure), indicating known locations of coesite and stishovite. They are only found in immediate association with thin pseudotachylite veins. See text for details. The limit of thermal metamorphism corresponds to the 200 °C post-impact isotherm. (After Martini, 1978, 1991.)

Langenhorst, 1994). Studies by TEM, however, indicate much smaller sizes for individual grains of coesite and stishovite, 1 μm–10 μm (White, 1993; Leroux *et al.*, 1994). The relatively large crystal sizes observed optically by Martini (1991) are polycrystalline aggregates in which the orientations of individual grains are only slightly different.

Vredefort is the only known case where coesite and stishovite are found in the parautochthonous rocks of the floor of a terrestrial impact structure. They occur outside the major thermal aureole that occurs in the parautochthonous rocks of the crater floor at Vredefort (Grieve *et al.*, 1990; Martini, 1991, 1992), and they are found only in association with thin pseudotachylite veins. They are not known from thicker pseudotachylite dikes, implying quenching is a necessary component for the formation and preservation of coesite and stishovite. They occur in areas where there is little obvious post-impact recrystallization but also where the shock pressures were apparently only sufficient to produce (0001) and {10$\bar{1}$3} PDFs in quartz (Grieve *et al.*, 1990). The overall paragenesis indicates recorded shock pressures generally lower than in most occurrences of stishovite and, particularly, coesite, at other terrestrial impact structures. It is likely that the high deviatoric stress-temperature gradient, which must have existed locally across the pseudotachylite-wall rock contact, played a role in localizing polymorph development (White, 1993). This may be analogous to the formation of ringwoodite from olivine in association with, and at the contact of, veins and dikes in some chondritic meteorites (Stöffler *et al.*, 1991).

Additional Aspects–We have focussed on the occurrence of planar microstructures and other phases of quartz as indicators of impact and, more importantly, sources of information on the recorded shock pressures at impact structures in non-porous targets. Several of the other analytical techniques for investigating the shock metamorphism of quartz have been applied to materials from impact structures in non-porous targets. Most of the studies, however, have been technique oriented or relatively non-systematic.

Reductions in refractive index of shocked quartz at a number of terrestrial impact structures are well documented (*e.g.*, see papers in French and Short, 1968; Engelhardt and Bertsch, 1969; Stöffler, 1971a, 1974). This technique, and its more elaborate form of measuring reflectivity, which was undertaken by Lambert (1981) on both experimentally shocked gneiss from the Rochechouart structure and naturally shocked anorthosite with diaplectic quartz glass from the Mistastin structure (Lambert and Grieve, 1984), has its limitations for systematic studies of naturally shocked polycrystalline rocks, because of shock reverberations and possible post-shock thermal effects. Refractive index changes are discussed in detail in Stöffler and Langenhorst (1994), where a number of other techniques are noted, such as x-ray, infrared absorption, electron paramagnetic resonance, x-ray photoelectron spectra, NMR and thermoluminescence, some of which have been applied to naturally shocked quartz from terrestrial impact structures in non-porous targets (*e.g.*, Fig. 21).

The conditions at impact structures are considerably less controlled and more complex than in shock recovery experiments. The differences that can exist between the recorded shock pressures in individual quartz grains in the same sample put severe limitations on all techniques that try to estimate average recorded shock pressures from observations on single constituent grains. Thus, any technique for pressure estimation must rely on some averaging procedure and may require considerable and, as yet unavailable,

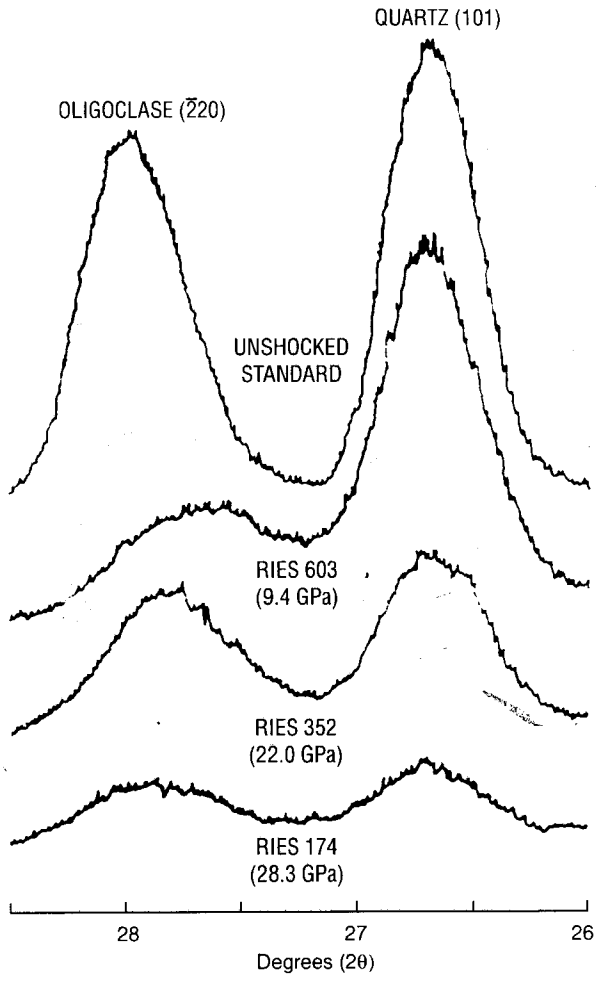


FIG. 21. X-ray diffractometer scan (0.2° 2Θ/min) of three shocked granitic rocks (pressure estimates in parenthesis) from the Ries structure, compared to unshocked standard. Note the reduction in peak heights (representing destruction of structural order) in quartz (and feldspar), with increasing recorded shock pressure. (After Hanss *et al.*, 1978.)

experimental calibration. This, in some sense, favors the application of the simpler techniques to estimate and study variations in the recorded shock pressure in quartz in the natural environment. The application of more sophisticated techniques, however, is still valid, both as an indicator of shock effects in quartz and for resolving problems related to the details and mechanisms of shock-induced changes in quartz (*e.g.*, Goltrant *et al.*, 1992; Langenhorst, 1994).

Impact Structures in Sedimentary (Porous) Targets

Shock Classification–The reaction of quartz to shock compression in porous sedimentary targets is distinctly different from that in non-porous crystalline targets, although the same principal types of shock effects occur. Due to the large difference in shock impedance between the solid grains and the pores in porous sedimentary targets, the shock wave energy is deposited very heterogeneously at the microscopic scale. This results in the coexistence of shock effects representative of very different shock pressures, which in non-porous rocks would generally occur in spatially separated lithologies. This requires a classification of shocked porous quartzose rocks and sediments in which the progressive stages of shock and the related pressure calibration is defined differently from crystalline rocks. Therefore, Fig. 1 and Tables 1, 3 and 4 cannot be applied. A well-documented shock classification has been developed by Kieffer (1971, 1975) and Kieffer *et al.* (1976). This classification is based on observations of shock effects in the Coconino sandstone at Barringer or Meteor Crater (Table 6). At the lowest pressure (~ <5 GPa), the porosity is reduced to zero and the constituent minerals are fractured. At moderate pressures (~5 GPa–13 GPa), fractured quartz coexists with minor amounts of glass and coesite (<20%). This assemblage is replaced at ~13 GPa–30 GPa by an increased fraction of glass and coesite, with traces of stishovite, and still abundant fractured quartz. At the highest shock levels, the main assemblage is glass (lechatelierite), coesite and minor amounts of fractured quartz. Planar microstructures are predominantly PFs oriented parallel to (0001) and $\{10\bar{1}1\}$. Compared to non-porous rocks, PDFs are rare, at least in the case of Barringer.

Planar Microstructures–To our knowledge, there have been no systematic studies of the spatial distribution of planar microstructures in quartz in parautochthonous lithologies at impact structures in porous sedimentary targets. There is also an additional complication in observations of planar microstructures in quartz at impact structures in sedimentary compared to crystalline targets. Robertson (1980), for example, noted a number of occurrences involving structures in sedimentary targets, where PDFs are rare, and others, where the $\{10\bar{1}3\}$ orientation is suppressed in favor of higher angle orientations, such as $\{11\bar{2}2\}$ and $\{10\bar{1}1\}$.

Barringer is the most striking example of a case where PDFs in quartz are rare. In samples of Coconino sandstone from allochthonous breccia, PFs are well developed (Bunch and Cohen, 1964), but PDFs occur in ~ <5% of the grains, although the samples may contain diaplectic quartz glass, coesite and stishovite (Kieffer, 1971). The PFs have orientations similar to those at impact structures in crystalline rocks. There is essentially no published information on the orientations of the relatively uncommon PDFs at Barringer except by Kieffer *et al.* (1976) who, in a TEM study, identified Brazil twins parallel to (0001) and glass lamellae, which they equated with "planar features." There are some data available, however, from quartz in sandstones at Steinheim (Table 7; Stöffler, 1972). Other examples where the development of PDFs is depressed include Wabar, where coesite is present (Chao *et al.*, 1961), but planar microstructures are restricted to only PFs (Bunch and Cohen, 1964; Short, 1966a), and Kamensk, where coesite has been observed but PDFs are apparently absent (Blank and Val'ter, 1983).

The second difference between planar microstructure development in crystalline and porous sedimentary targets is exemplified at the B.P. and Oasis structures, where PDFs are better developed in the sandstone target rocks than at Barringer. Although only histograms of pole orientations are available (French *et al.*, 1974) and it is not possible to reconstruct the frequency percentage of uniquely indexed PDF planes, the distribution of PDF orientations is clearly different from those observed in quartz from crystalline targets. Although apparently present, $\{10\bar{1}3\}$ and $\{10\bar{1}2\}$ orientations are subordinate to higher angle orientations, such as $\{11\bar{2}2\}$, $\{10\bar{1}1\}$ and $\{51\bar{6}1\}$ (Fig. 22). Concussion fractures and planar fractures, such as those observed in the Coconino sandstone (Kieffer, 1971), are also a common characteristic of shocked quartz grains at these structures (French *et al.*, 1974). A similar situation is also

TABLE 6. Progressive shock metamorphism of porous sandstone and pressure-shock temperature conditions based on observations of Coconino sandstone, from Stöffler (1984) and modified after Kieffer *et al.* (1976).

Shock* class	Pressure** GPa	Shock temperature °C	Shock produced phenomena
0	0.2–0.9	~25	Undeformed sandstone
Ia	~3.0 (2.2–4.5)	~250	Deformed sandstone with remnant porosity
Ib	~5.5 (3.6–13)	~350	Deformed sandstone compressed to zero porosity
II			Dense sandstone with 2%–5% coesite, 3%–10% glass and 80%–95% quartz
III	~13	~950	Dense sandstone with 18%–32% coesite, traces of stishovite, 0%–20% glass and 45%–80% quartz
IV	~30	>1000	Dense sandstone with 10%–30% coesite, 20%–75% glass and 15%–45% quartz
V			Vesicular (pumiceous) rock with 0%–5% coesite, 80%–100% glass (lechatelierite) and 0%–15% quartz

* Note shock classes and their pressure ranges do not correspond to the shock stages for crystalline quartzo-feldspathic rocks (Table 4).
** Average pressure; locally, peak pressures can be much higher; extreme pressure estimates are given in parentheses (see Kieffer *et al.*, 1976).

observed at Steinheim (Table 7) and Avak and Tookoonooka (Fig. 23; Gorter *et al.*, 1989), where the dominant PDF orientations are {11$\bar{2}$1}, {10$\bar{1}$0} and other high angle planes.

Gosses Bluff is an intermediate case. Planar fractures and PDFs are common in quartz from sandstones at Gosses Bluff, and Crook and Cook (1966) have produced a histogram of angular data on PDFs (Fig. 24), similar to that from B.P. and Oasis (Fig. 22). On the other hand, Milton *et al.* (1972) noted that, when there are few PDFs, (0001) orientations dominate and that, where they are more abundant, {10$\bar{1}$3} orientations dominate. This is in accord with Geological Survey of Canada data, which indicate that PDF orientation distributions at Gosses Bluff are more similar to those from crystalline targets (Fig.12) than from structures such as B.P., Oasis, Avak and Tookoonooka (Fig. 23). Indeed, in some cases, the relative abundances of the various orientations of the PDF at craters in sedimentary targets are equivalent to those found at craters in crystalline targets. For example at Sierra Madera, reconstructing the frequency percentages of PDFs in quartz in sandstone clasts in breccias, which were indexed directly (Wilshire *et al.*, 1972), and comparing them with similar observations from crystalline clasts from the Ries suevite (Engelhardt and Bertsch, 1969) or parautochthonous target rocks at West Clearwater (Engelhardt *et al.*, 1968) produces essentially equivalent results (Fig. 25; Table 7). Other examples of craters in sedimentary targets that have PDF orientations apparently equivalent to those in crystalline targets include Azuara (Ernston *et al.*, 1985), Chilyi (Vishnevsky, 1989) and Crooked Creek (Dietz and Lambert, 1980).

Although these observations in sedimentary targets were all made on sandstones, the cause of the variation in recorded PDF orientations appears to be due to differences in texture (*e.g.*, porosity and grain size distribution). Milton *et al.* (1972) noted that at Gosses Bluff, PDFs are more common in silicified sandstone than in adjacent more porous sandstone. Thin sections of sandstones from Gosses Bluff held at the Geological Survey of Canada in which the distribution of PDF orientations is similar to crystalline targets (*cf.*, Figs. 23 and 25) have a texture of interlocking quartz grains of similar size. Similarly, although details are variable, it appears that the lithologies at Azuara, Chilyi, Crooked Creek and Sierra Madera are relatively equigranular sandstones with tightly interlocking grains. On the other hand, the sandstone with PDFs at Gosses Bluff, illustrated by Crook and Cook (1966) (and presumably an example of the lithology in which their orientations were measured), shows relatively large rounded quartz grains set in a finer matrix of smaller rounded quartz grains. Illustrations of the sandstone target rocks at B.P. and Oasis indicate larger (>0.5 mm) grains of subrounded quartz in a porous matrix of angular fragments of quartz (see Fig. 2 in French *et al.*, 1974). The Coconino sandstone at Barringer forms an end member to this textural progression, having an average porosity of ~ <10% but reaching ~ <20% locally (Kieffer, 1971). The exact effects of variations in porosity and grain size (morphology) distribution are not well known at present and are the subject of current study (*e.g.*, Grieve and Therriault, 1995).

One suggestion for the apparent greater abundance of {11$\bar{2}$2} and {10$\bar{1}$1} PDFs in some sedimentary targets, which generally reflect higher recorded shock pressures than {10$\bar{1}$3} orientations (Robertson and Grieve, 1977; Stöffler and Langenhorst, 1994) is that lower pressure (0001) and {10$\bar{1}$3} PDFs do not develop because at lower shock pressures the pressure-volume work done on the rocks is absorbed in closing existing pores and voids. It is only at higher pressures, when the rocks are compacted, that PDFs develop, and their orientations, thus, reflect these higher pressures (Robertson, 1980). Observations at Barringer on the Coconino sandstone demonstrate compaction of pores and plastic deformation of quartz grains to produce an interlocking array of quartz grains with increasing recorded shock pressure (Kieffer, 1971). It has been demonstrated that porosity has a significant

TABLE 7. Absolute frequency percentage of planar microstructures in quartz in some samples from Ries (crystalline) and Sierra Madera, Steinheim and Barringer (sedimentary).

Miller Index	Ries	Sierra Madera	Steinheim	Barringer
(0001)	3	11	34	29
{10$\bar{1}$3}	45	45	9	0
{10$\bar{1}$2}	22	25	3	0
{10$\bar{1}$1}	13	11	29	39
{11$\bar{2}$2}	3	3	4	22
{11$\bar{2}$1}	2	0	3	0
{21$\bar{3}$1}	4	0	1	0
{51$\bar{6}$1}	4	0	5	0
{10$\bar{1}$0}	3	4	2	9
{11$\bar{2}$0}	3	4	2	9

Data for Ries from Engelhardt and Bertsch (1969); for Sierra Madera, from Wilshire *et al.* (1972); for Steinheim, from Bertsch (1968) and Stöffler (1972); and for Barringer, from Bunch and Cohen (1964). See text for details and Engelhardt and Bertsch (1969) on calculation of absolute frequency percentage. Note that the data for Sierra Madera are for PDFs; for Ries and Steinheim, a mixture of PDFs and PFs; and for Barringer, only PFs.

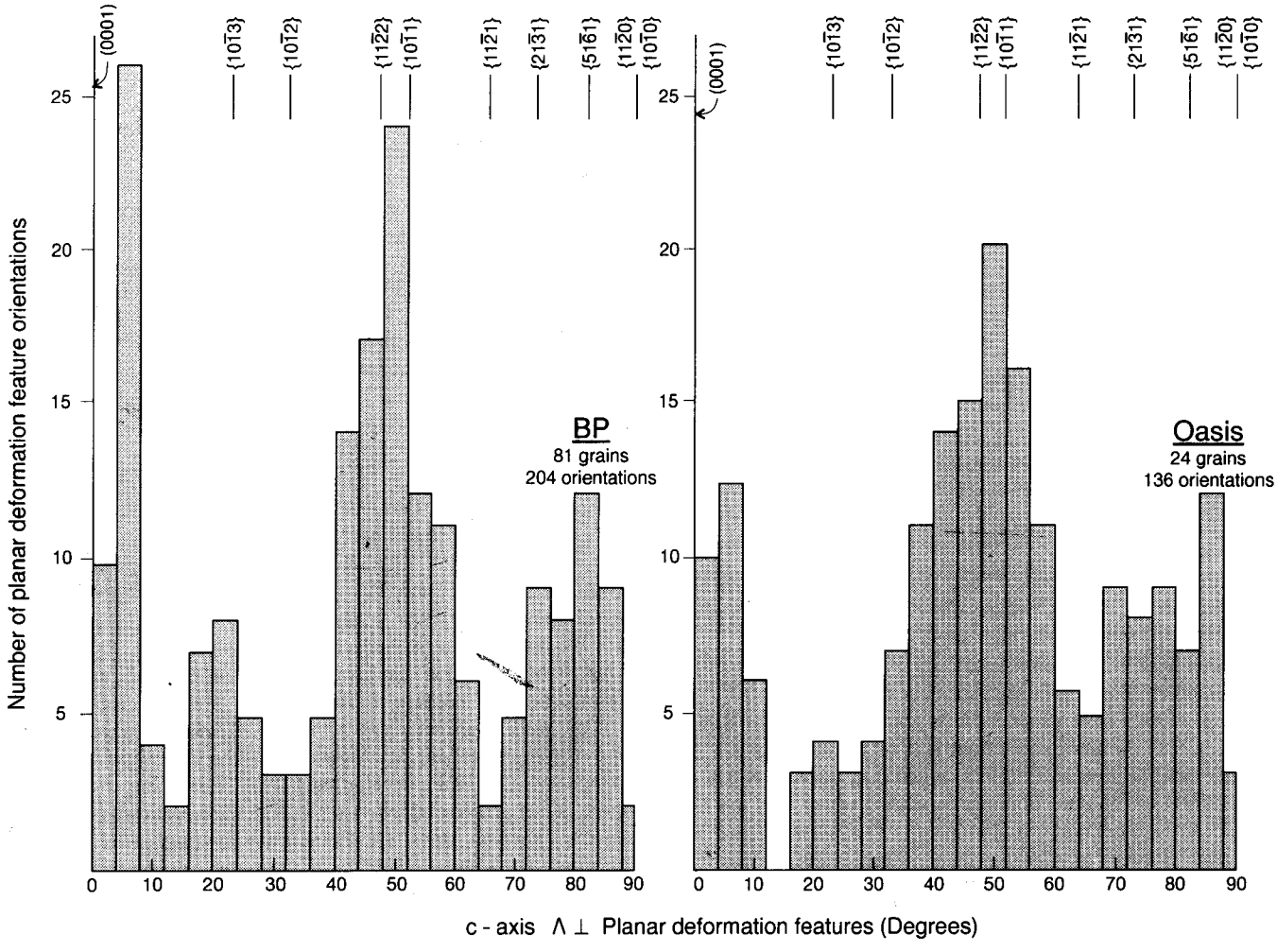


FIG. 22. Histogram of angles, binned at 4°, between c-axis and poles to PDFs in quartz from sandstones at the B.P. and Oasis structures. Note the relatively low abundance of {10$\bar{1}$3} and {10$\bar{1}$2} (*cf.*, Fig. 11) orientations, compared to higher angle orientations, such as {11$\bar{2}$2}, {10$\bar{1}$1} and others. Data are from French *et al.* (1974). See text for details. Note that this is a mixture of quartz grains recording different degrees of shock.

effect on the local pressure-temperature conditions achieved during shock metamorphism (Kieffer *et al.*, 1976).

Diaplectic Quartz Glass, Lechatelierite, Coesite and Stishovite–Reports of diaplectic quartz glass from impact structures in porous sedimentary targets are relatively rare and always in samples from allochthonous materials (*e.g.*, Masaitis *et al.*, 1980). Due to the complications of shock reverberations in porous lithologies (Kieffer, 1971; Kieffer *et al.*, 1976), individual quartz grains tend to melt at the margins of pores, as the result of high post-shock waste-heat localized in the pores. Thus, glassy phases occur in very complex relationships in shocked porous quartzose rocks and sediments. According to Kieffer *et al.* (1976), three types of SiO_2 glass are formed locally from quartz grains in the Coconino sandstone at Barringer: (1) diaplectic glass occurs in shock classes II, III and IV (Table 6) as lamellae or domains in quartz and coesite; (2) "froth", with extremely small, tens of Angstroms wide, angular vesicles forms thin layers (~1 μm) on host quartz grains of class III rocks near to the regions, where high pressure phases are formed (former pores); and (3) lechatelierite with larger, spherical vesicles is found in areas of the size of original quartz grains, predominantly in class IV rocks. Fluidal lechatelierite is the result of mineral melting (Table 6; shock class V). This highly vesiculated glass contains relict quartz clasts and coesite.

Coesite (Chao *et al.*, 1960) and stishovite (Chao *et al.*, 1962) were first discovered as naturally occurring minerals in samples from the Coconino sandstone at Barringer. In moderate to strongly shocked samples from allochthonous breccia, coesite can constitute > ~30% of the sample (Table 6; Kieffer, 1971). Trace amounts of stishovite (<1%) also occur in these samples (Fahey, 1964; Kieffer, 1971). Stishovite, however, disappears, and the amount of coesite decreases in the most strongly shocked rocks, which consist essentially of vesiculated lechatelierite and remnant fragments of quartz (Kieffer, 1971). Coesite and stishovite are particularly prevalent in samples from the fall-back layer at Barringer. Although the fall-back layer contains samples from all shock classes, all the samples of shock class III and above (Table 6), examined by Kieffer (1971) and Kieffer *et al.* (1976), are believed to be from the fall-back layer. High contents of coesite (20%) occur also in sandstone clasts in breccias at the Kamensk structure (Blank and Val'ter, 1983). These figures can be compared with coesite

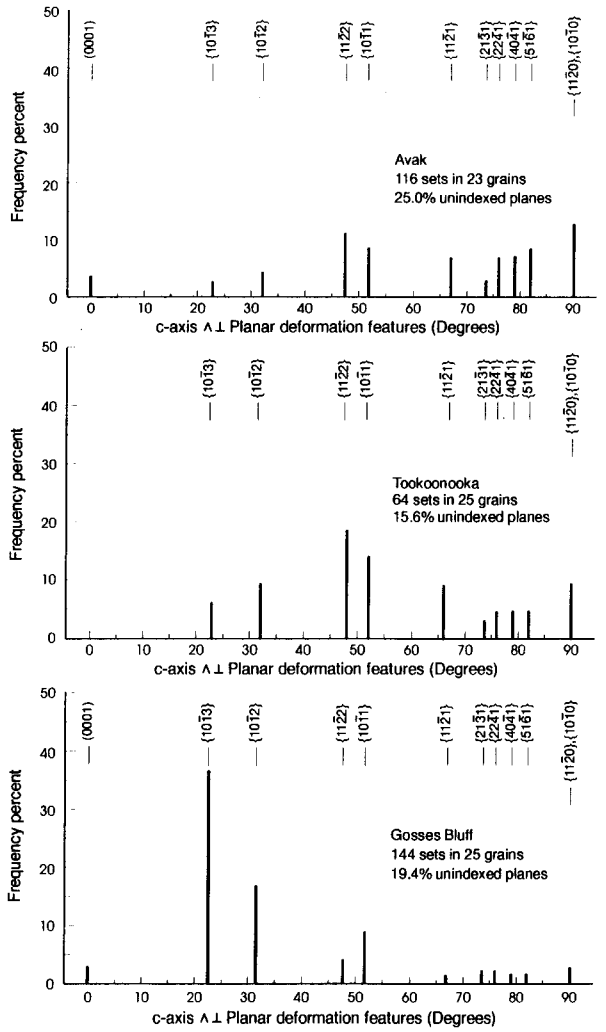


FIG. 23. Histogram of frequency percent of uniquely indexed PDF orientations, using a template containing known planar microstructure orientations, in quartz from the Avak, Tookoonooka and Gosses Bluff structures. Note the prevalence of high angle orientations at Avak and Tookoonooka, which is similar to Fig. 22 and in contrast to Fig. 12 and Gosses Bluff. Note that this is a mixture of quartz grains recording different degrees of shock. Data are from Geological Survey of Canada.

replacing up to ~40% and stishovite up to ~5%, but mostly <1%, of the original quartz in shocked crystalline clasts in suevite at the Ries (Stöffler, 1971a). As quartz is only one component of these crystalline rocks at the Ries, the percentage of high-pressure polymorphs of quartz for the samples, as a whole, is much less. A detailed account of the processes occurring during shock compression and subsequent rarefraction in porous rocks is given in Kieffer (1971) and Kieffer *et al.* (1976). As exemplified by the response of the Coconino sandstone to shock, it is a complex process involving the physical collapse of pores, jetting phenomena with the extrusion of molten silica into pores, the formation and reversion of stishovite, the crystallization of various morphological types of coesite and other complex phenomena.

Additional Aspects–We have focussed, to some degree, on the development and orientation of planar microstructures in quartz from impact craters in porous sedimentary targets, as there are differences with respect to equivalent shock features in crystalline

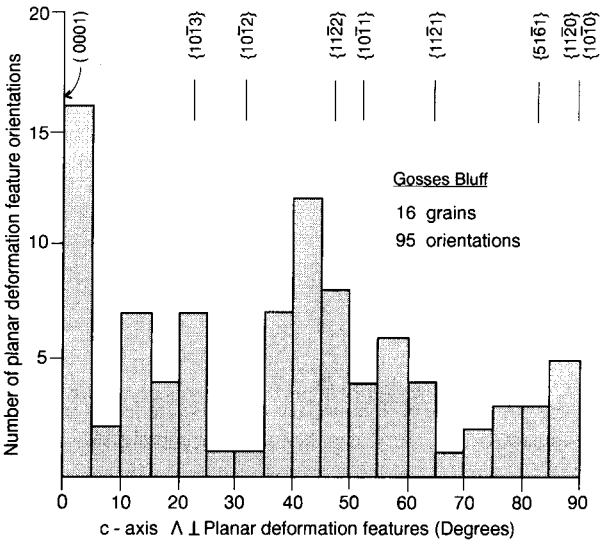


FIG. 24. Histogram of angles between c-axis and poles to PDFs, binned by 5°, in quartz in sandstones from the Gosses Bluff structure. Note the similarity to distributions from B.P. and Oasis (Fig. 22) and difference from other data from Gosses Bluff (Fig. 23). See text for full discussion. Note that this is a mixture of quartz grains recording different degrees of shock. (After Crook and Cook, 1966.)

targets. Other techniques have been used to characterize shocked quartz from porous targets. For example, x-ray asterism has been studied in quartz from Barringer and Wabar craters, with the results being essentially equivalent to those from crystalline targets (Dachille *et al.*, 1968). An interesting application at Barringer has been the use of thermoluminescence. Sutton (1985a) studied the shock dependence of the thermoluminescence properties of quartz from the Coconino sandstone and noted that not only did the thermoluminescence sensitivity decrease with increasing recorded shock pressure but was essentially reset by shock pressures >13 GPa. In samples in which the thermoluminescence was reset by the impact event, Sutton (1985b) used the current thermoluminescence and post-impact radiation to calculate an age for the Barringer impact event of 49 000 ± 3000 years.

Studies by NMR spectroscopy of five samples of Coconino sandstone indicate peak width increases with increasing recorded shock pressures, as judged by the pressure scheme of Kieffer (1971), and the development of coesite, stishovite and finally only coesite (Boslough *et al.*, 1993; Cygan *et al.*, 1994). Cygan *et al.* (1994) also discovered what appears to be a hydroxolated amorphous silica phase. This phase, however, could not be found in experimental shock products of wet Coconino. Similarly, NMR peaks corresponding to coesite and stishovite were not observed in the experimental products (Cygan *et al.*, 1994), presumably reflecting the different pressure-temperature-time histories of experimentally and naturally shocked samples.

Distal Ejecta

Some of the previous discussion has dealt with examples of shock metamorphic effects in quartz from suevite breccia at the Ries structure and with the fall-back breccia at the Barringer structure. Little or no distinction was made as to whether the suevite occurred within the structure or as part of the once continuous ejecta outside the structure. Here, however, we consider only ejecta that is well removed from the impact site, highly dispersed and quickly cooled.

Disregarding tektite and microtektite occurrences, there are only two currently known examples, where such distal ejecta are linked to a known impact site: (1) ejecta from the Acraman structure, which occur up to ~500 km from the impact site (Gostin *et al.*, 1986) and (2) ejecta from the Chicxulub structure, which occur world wide at the Cretaceous-Tertiary (K/T) boundary (*e.g.*, Bohor *et al.*, 1984,1987; Hildebrand *et al.*, 1991).

In the case of Acraman, the discovery of a layer containing shocked quartz in dacite clasts in the shales of the Adelaide geosyncline, South Australia, was the initial evidence for ejecta from an impact structure. Later work indicated an associated siderophile element anomaly (Gostin *et al.*, 1989; Wallace *et al.*, 1990). There is very little documentation of the nature of the planar microstructures, except that $\{10\bar{1}3\}$ PDFs dominate over $\{11\bar{2}2\}$ and $\{22\bar{4}1\}$ orientations (Gostin *et al.*, 1986).

In the case of the K/T boundary deposits, the initial indications of an impact origin were geochemical in nature (Alvarez *et al.*, 1980; Ganapathy, 1980). This was followed by the discovery of physical evidence of impact in the form of quartz with PDFs in K/T boundary samples from the western interior of North America (Bohor *et al.*, 1984). In addition to PDFs, some quartz grains in the K/T boundary layer display reduced refractive indices and x-ray asterism (Bohor, 1990). Since the initial discoveries, quartz with PDFs and other shocked minerals have been recognized world wide in K/T boundary deposits (Bohor *et al.*, 1987). Surprisingly, little systematic work has been done on the orientation of PDFs in quartz from K/T boundary deposits. Bohor *et al.* (1987) illustrated histograms of angles to the c-axis for 47 grains from five K/T sites world wide. The dominant orientations appear to be $\{10\bar{1}3\}$ and $\{10\bar{1}2\}$, although they did not index the planes directly. Grieve and Alexopoulos (1988), indexing the PDFs directly, obtained a similar result for 35 grains from one K/T site in Alberta, Canada. Izett (1987) measured the orientations of 200 PDFs from K/T boundary deposits in the Raton Basin, U.S.A., and concluded that over 50% of the angles to the c-axis correspond to the $\{10\bar{1}3\}$ orientation. Badjukov *et al.* (1986) produced a histogram of angular data on PDF orientations in quartz from two K/T sites in the former Soviet Union, which also suggest $\{10\bar{1}3\}$ as the dominant orientation.

The most notable feature of the distribution of angular data on PDFs is that almost no basal features were apparently observed optically (Fig. 26; Izett, 1987; Bohor *et al.*, 1987; Grieve and Alexopoulos, 1988). At the level of the optical microscope, the PDFs are generally not decorated. At the scale of the TEM,

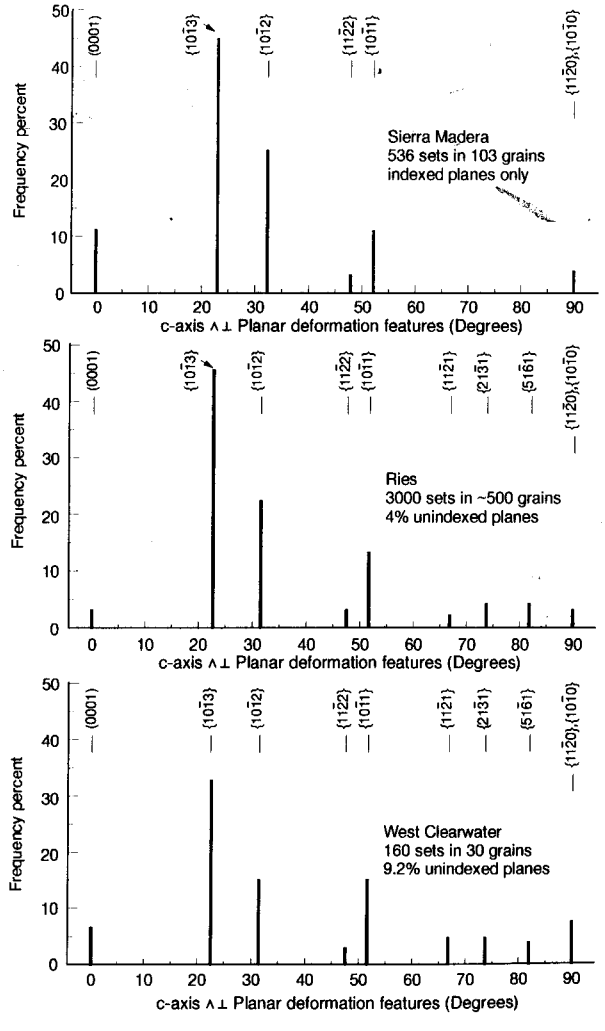


FIG. 25. Histogram of frequency percent of uniquely indexed PDFs, using a template containing known planar microstructure orientations, in quartz from sandstones from Sierra Madera (Table 8). Note the differences with other samples and sedimentary targets (Fig. 23) and closer similarity to the results from quartz from impact structures in crystalline targets (Fig. 12), Ries (data in Table 8) and W. Clearwater (data from Engelhardt *et al.*, 1968). Also note that this is a mixture of quartz grains recording different degrees of shock.

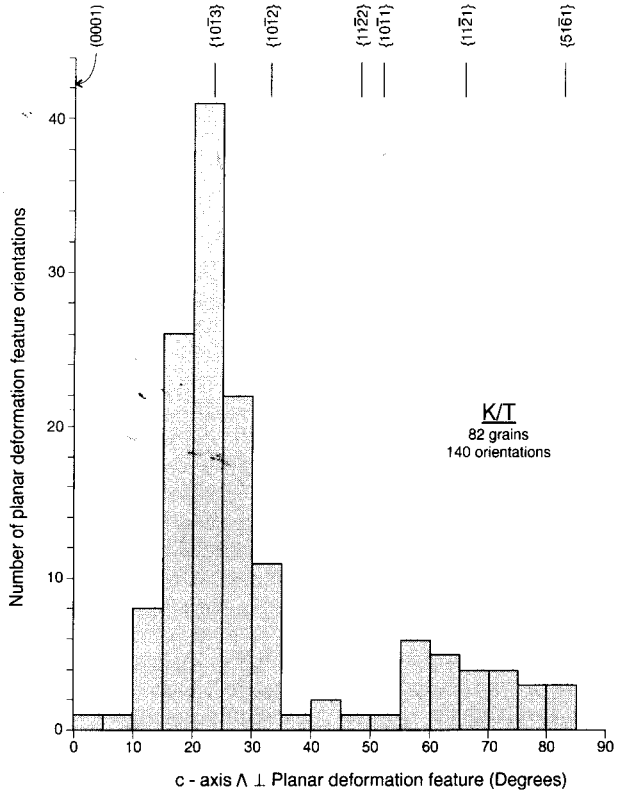


FIG. 26. Histogram of angles between c-axis and poles to PDFs, binned by 5°, in quartz from North American K/T boundary sediments (Raton Basin, U.S.A.; Scollard Canyon, Canada). Data are from Bohor *et al.* (1987) and Grieve and Alexopoulos (1988). Note that this is a mixture of quartz grains recording different degrees of shock.

however, some of the PDFs with {10$\bar{1}$3} and {10$\bar{1}$2} orientations contain a high density of tiny bubbles, indicating some post-shock, water-assisted recrystallization (Goltrant *et al.*, 1991, 1992). The PDFs that consist of thin lamellae of amorphous silica, however, are also observed (Gratz *et al.*, 1992b; Goltrant *et al.*, 1992), as well as (0001) Brazil twins (Leroux *et al.*, 1995). The nonrecrystallized to minor recrystallized nature of the PDFs in quartz at the K/T boundary sites examined is the reason for the relative absence of basal PDFs in the optical data. We have recently reexamined some shocked quartz grains from two K/T boundary sites in North America and observed optically only a very small component (~2%) of basal (0001) PDFs (Fig. 27).

No systematic study has been undertaken to determine whether there is any variation in the orientation of PDFs and, thus, recorded shock pressure with location in quartz from K/T boundary layers. The maximum dimensions of the shocked quartz grains, however, tend to be larger, and there are more grains containing PDFs at North American sites (Bohor and Izett, 1986), suggesting an impact site close to the Americas. This has since been confirmed with the discovery of the Chicxulub impact structure in the Yucatan (Hildebrand *et al.*, 1991). Bohor *et al.* (1984) claimed to have detected faint x-ray diffraction lines corresponding to stishovite in shocked quartz from the K/T deposits in the Raton Basin, but their findings could not be duplicated by Izett (1987). McHone *et al.* (1989), however, detected stishovite in K/T samples from the Raton Basin by NMR. They also obtained x-ray diffraction data that indicated eight peaks that could be indexed to stishovite. Most recently, the occurrence of coesite in K/T boundary materials has been identified by NMR and confirmed by x-ray diffraction (Boslough *et al.*, 1995).

There have been claims of quartz containing PDFs at other biostratigraphic boundaries (*e.g.*, at the Triassic-Jurassic boundary; Badjukov *et al.*, 1987; Bice *et al.* 1992). The observation basis for these claims, however, is not as strong as for the K/T boundary event and the observations are somewhat confusing, with the "shocked" quartz occurring in three separate beds (Bice *et al.*, 1992). Recently, however, shocked quartz with PDFs has been recognized near the Eocene-Oligocene boundary at Massignano, Italy (Langenhorst and Clymer, 1995).

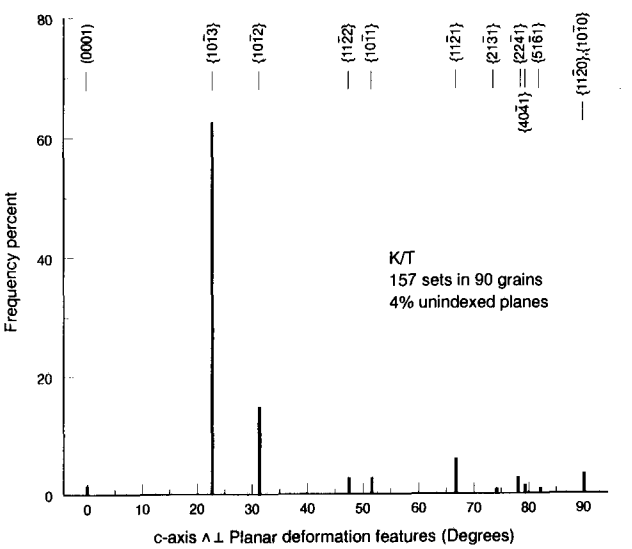


FIG. 27. Histogram of frequency percent of uniquely indexed PDFs, using a template containing known planar microstructure orientations, in quartz from North American K/T boundary sediments (Raton Basin, U.S.A., Scollard Canyon, Canada). Note the very small frequency of occurrence for basal (0001) PDFs. See text for details. Note that this is a mixture of quartz grains recording different degrees of shock. Data are from the Geological Survey of Canada.

Quartz grains with PDFs and coesite have been identified in the upper portion of the North American microtektite layer in DSDP site 612 (Glass, 1989), site 149, Piston Core RC9-58 in the Caribbean, site 94 in the Gulf of Mexico and Barbados (Glass and Wu, 1992). Coesite has also been identified in at least six sites of the Australasian strewn field, with the highest concentrations of coesite occurring in sites close to Indochina (Glass and Wu, 1992). Systematic studies of the shock effects have not been undertaken and the source craters for both microtektite strewn fields are unknown, although a number of sites have been suggested. Coesite is also known as inclusions in some tektites (Walter, 1965).

Nuclear Explosions

Although nuclear explosions are technically not natural occurrences, we include them here for completeness and because they form a bridge between the different temporal and post-shock thermal regimes of experiments and natural impact events. Short (1966b, 1968) examined the shock metamorphism of quartz from the Climax granodiorite stock after the 5 ± 1 kiloton Hardhat explosion. The samples were parautochthonous materials and the planar microstructures were nondecorated (Fig. 28), with the most prominent peaks in the histogram of angles relative to the c-axis corresponding to PDF orientations {10$\bar{1}$3} and {10$\bar{1}$2}. A "few" basal features were observed but Short (1968) was equivocal as to whether they were PFs or PDFs. At estimated shock pressures >30 GPa, there were indications of diaplectic glass but the sample had been subjected to 1000 °C for several hours due to the thermal radiation and had been recrystallized. Similar observations were made by Borg (1972) at the 61 ± 10 kiloton Piledriver explosion in the same granodiorite. Some quartz grains developed planar microstructures at calculated shock pressures of ~7.5 GPa. The number of sets of PDFs increased with shock pressure and the most common orientation was {10$\bar{1}$3}, with subordinate {10$\bar{1}$1} and {10$\bar{1}$2}. Subbasal (<10°) features occurred, with a frequency of <6%, but no comment was made as to their nature (Borg, 1972). Neither Borg (1972) nor Short (1966b, 1968) reported the occurrence of high-pressure polymorphs; however, coesite has been identified optically

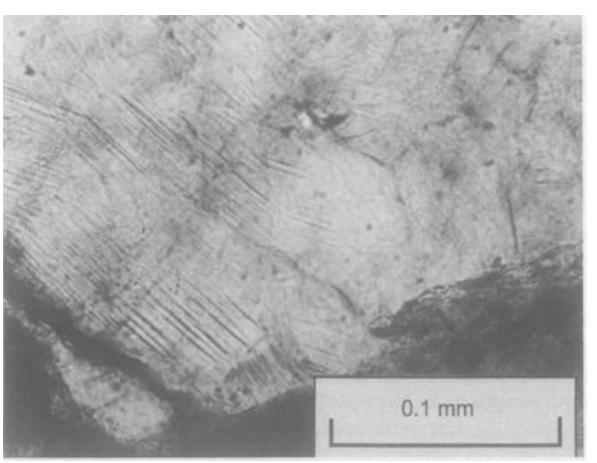


FIG. 28. Photomicrograph (crossed polars) of undecorated PDFs in quartz in granodiorite as the result of the Hardhat nuclear explosion.

in material from Hardhat (Stöffler, 1971a). Optical data from the Geological Survey of Canada for PDFs in quartz from Hardhat and Piledriver are shown in Fig. 29.

The ejecta from the 100 kiloton Sedan explosion was examined by Short (1968, 1970b) for shock metamorphic effects. In granodiorite boulders from the alluvium, his observations were essentially the same as for shocked granodiorite from Hardhat. The only difference was that some samples contained diaplectic quartz (and feldspar) glasses. Presumably, their occurrence was due to the samples being ejecta and, thus, not subjected to the residual thermal pulse experienced by the parautochthonous granodiorite samples at Hardhat (Short, 1968). There were also samples of strongly cemented orthoquartzite boulders in the ejecta. As with the granodiorite samples, PDFs and PFs were developed. The PDFs were nondecorated with $\{10\bar{1}3\}$ as the dominant orientation. Other orientations included $\{10\bar{1}2\}$, $\{11\bar{2}2\}$, $\{10\bar{1}1\}$, $\{11\bar{2}1\}$ and $\{51\bar{6}1\}$. The PDFs with $\{10\bar{1}3\}$, $\{10\bar{1}2\}$ and, rarer $\{10\bar{1}1\}$ orientations have been identified in a recent TEM study, which also indicated incipient recrystallization of some PDFs (Cordier and Gratz, 1995). Diaplectic quartz glass and lechatelierite were observed in some samples and, based on x-ray identification of the two main reflections, one highly-shocked and vesiculated sample contained coesite (Short, 1970b). Dachille *et al.*, (1968) also detected coesite in samples from Sedan. Some basal features were noted in one sample, but it was not resolved whether they were PFs or PDFs. In addition to the optical observations, Short (1970b) noted increasing x-ray asterism, reduced refractive indices, the occurrence of biaxial optical figures for quartz, decreasing peak heights in thermoluminescence spectra and loss of fine structure in infrared adsorption spectra with increasing shock metamorphism as defined on the basis of optical observations.

In summary, although detailed observations of the shock metamorphism of quartz from nuclear (and TNT, which are not discussed here) explosions are somewhat limited, they are consistent with those at terrestrial impact structures (Short, 1968). The PDFs are generally non- or only weakly-decorated (Fig. 28) due to the absence of long-term annealing effects (Cordier and Gratz, 1995), and their orientations (Fig. 29) are equivalent to those in natural impact events into similar rock types. Some question remains, however, as to the occurrence of features with a basal orientation. They are rare and the literature is not clear as to their exact nature (Short, 1968).

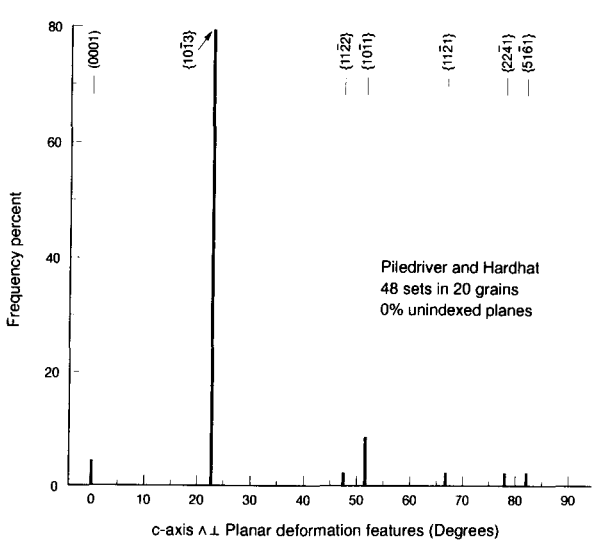


FIG. 29. Histogram of frequency percent of uniquely indexed PDFs, using a template containing known planar microstructure orientations, in quartz from two samples from Piledriver and Hardhat nuclear explosions, respectively. Note the rare occurrence of basal (0001) features. The absence of $\{10\bar{1}2\}$ features indicates that the recorded shock pressure is <20 GPa in the two samples. Data are from the Geological Survey of Canada.

COMPARISON OF SHOCK-METAMORPHOSED AND ENDOGENICALLY DEFORMED QUARTZ

There have been a number of recent optical studies focussing on quartz containing planar microstructures from volcanic and tectonic sites (*e.g.*, Lyons *et al.*, 1993). These studies have led to some confusion, in particular concerning the significance of PDFs as an unique shock indicator. It is inherent in the definition and physical nature of shock waves that strain rates exceed 10^{-5} s^{-1}, and the pressure duration is of the order of nanoseconds (laboratory) to seconds (the largest natural impacts). Typical strain rates for tectonic deformation range from ~10^{-12} s^{-1} to 10^{-14} s^{-1} with the time span of tectonic compression normally of the order of 10^4 years to 10^6 years. The time span of tectonic phenomena reaches short times, possibly minutes, only in faults and landslides (Spray, 1992). Similarly, there is no physical basis for the production of strong shock waves by volcanic explosions. Volcanic "explosions" are sustained decompression events (de Silva *et al.*, 1990; Gratz *et al.*, 1992b), with maximum overpressures of <0.5 GPa (Self *et al.*, 1979; Wilson *et al.*, 1980). Considering the totally different physical conditions in volcanic explosions, it is unrealistic to assume that volcanic explosions could produce isotropic lamellae, such as PDFs. Even static high-pressure experiments have shown that the isotropization of quartz requires at least pressures of >1.5 GPa at 600 °C (Dell'Angelo, 1993), with complete transition requiring ~25 GPa–30 GPa at room temperature (Cordier *et al.*, 1993; Kingma *et al.*, 1993). Such conditions are clearly not fulfilled in volcanic events. For these reasons, it is incorrect and very confusing, when rare deformation lamellae in volcanic tuffs are called "shock lamellae" and to speak of "shock mosaic structure" (*e.g.*, Carter *et al.*, 1986, 1990; Rice, 1987; Lyons *et al.*, 1993).

Engelhardt and Bertsch (1969) briefly reviewed the optical character of the deformation and transformation of quartz by endogenic processes compared to shock effects resulting from impact. Their conclusions are still valid. Since then, considerable additional data have been acquired, particularly through TEM studies, which confirm that the planar microstructures produced by shock can be distinguished unequivocally from all other types of natural deformation or transformation effects. For example in a recent TEM study, Cordier *et al.* (1994) have demonstrated that microstructures in quartz from the Susice and Sevetin structures, which were previously suggested to be PDFs (*e.g.*, Vrána, 1987), are in fact numerous subparallel sub grain boundaries, and these structures are not impact in origin.

Two types of permanent effects in quartz resulting from endogenic processes or from static high-pressure laboratory experiments can be distinguished: (1) plastic deformations and (2) phase transformations. Plastic deformation of quartz is common in crustal rocks and recognized in the microscope as undulatory extinction. The intensity of this deformation is variable and, if severe, may grade into mosaicism. In contrast, shock compression

typically induces very strong mosaicism, which is quantitatively different from that of non-shock environments, as shown by optical and x-ray studies (*e.g.*, Dachille *et al.*, 1968; Hörz and Quaide, 1972; Schneider *et al.*, 1984). Tectonic plastic deformation may result in deformation bands and deformation lamellae, as defined and described, for example, by Christie and Raleigh (1959). Zones of undulatory extinction or deformation bands are often perpendicular to deformation lamellae. In contrast to deformation bands, which are elongated and not strictly planar zones, deformation lamellae (Böhm lamellae) are planar, subplanar or wave-shaped lamellae that are arranged in parallel to subparallel sets (Fig. 30). In strong contrast to shock-induced planar microstructures (particularly PDFs), one set per grain is typical while two sets of different orientations are very rare (Table 8). The lamellae, commonly >2 μm in width, consist either of minute inclusions or cavities arranged in planar zones, or they represent lamellae with slightly different optical orientation and refractive index from the host quartz. They are asymmetric, having slightly higher refractive indices and birefringence on one side (Carter and Friedman, 1965). None of these optical properties pertain to shock-induced planar microstructures (Table 8). Most importantly, however, their crystallographic orientation is completely different from the orientation of shock-induced PDFs. They are not generally parallel to specific rational low indices planes and the range of orientations indicates no evidence for preferred crystallographic orientation, except that they are mostly inclined to the c-axis by ~10° to 30° (Fig. 31).

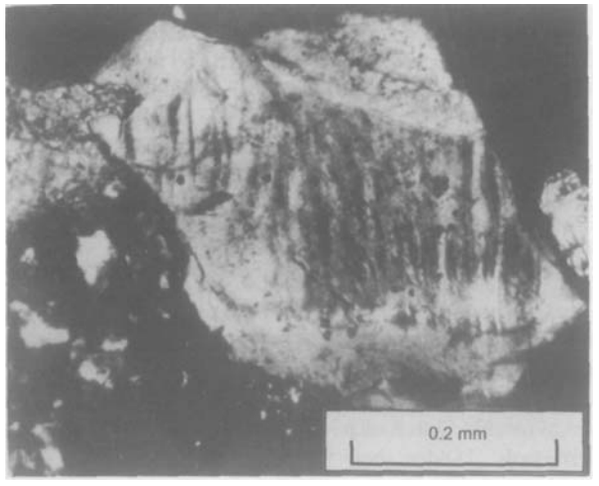


FIG. 30. Photomicrograph (crossed polars) of tectonic Böhm lamellae in quartz from Dry Creek Ridge anticline, Montana, U.S.A. Photomicrograph supplied by A. R. Huffman.

It is also well known from TEM investigations that the microstructure of deformation lamellae in tectonically deformed quartz differs from that of shock-produced PDFs (McLaren and Hobbs, 1972; White, 1971, 1973; Liddell *et al.*, 1974; Christie and Ardell, 1974, 1976). As indicated by microstructural observations (see review in McLaren, 1991), deformation lamellae represent parallel, closely spaced (spacing about a few micrometers) subgrain boundaries consisting of well-ordered networks of two or three sets of dislocations. Deformation lamellae, as well as healed cracks, have also been documented by TEM from high strain rate tectonic environments (*i.e.*, in fault zones; Blenkinsop and Drury, 1988). In contrast to deformation lamellae, fresh PDFs, with the exception of basal orientations, are amorphous SiO_2 lamellae (Table 8). In addition, shock recovery experiments have shown that dislocations cannot be induced by shock compression, due to its short time scale and extremely high strain rates (Gratz *et al.*, 1992a; Langenhorst, 1994). Deformation lamellae (due to tectonic deformation) and their host quartz are characterized in TEM studies by very high dislocation densities (Table 8).

Deformation lamellae, deformation bands and undulatory extinction have been reproduced in static high-pressure experiments at elevated temperatures of >500 °C (*e.g.*, Christie *et al.*, 1964a; Carter *et al.*, 1964). Their properties and crystallographic orientations are very similar to those naturally occurring deformation lamellae (Fig. 31). At low temperatures, planar fractures ("faults") can be produced by static compression (Christie *et al.*, 1964b), as a result of brittle failure. These faults form parallel sets of extension fractures parallel to (0001), $\{10\bar{1}1\}$ and $\{01\bar{1}1\}$, which are phenomenologically similar to shock-produced PFs, but do not coexist with deformation lamellae, which form by plastic deformation.

TABLE 8. Comparison of characteristics of deformed quartz and other phases of silica resulting from impact into crystalline targets and in endogenic geologic processes.

Property	Shock-metamorphic rocks	Endogenically deformed and metamorphosed rocks
Quartz with planar microstructures		
Percentage of deformed grains	≤100%	<30%
Sets per grain	1–15	1–2
Crystallographic orientation	Specific rational low-indices planes (*e.g.*, $\{10\bar{1}3\}$, $\{10\bar{1}2\}$)	Broad variation, majority subparallel to (0001) to ~30° to c-axis
Morphology	Planar	Often subplanar or bent
Structural state	Amorphous (in the case of all but (0001) PDFs)	Crystalline
Optical state of host grain	Mosaicism	Undulatory extinction and deformation bands
Structural state of host grain	Small scale domains but no enhanced dislocation density	Typically very high dislocation density
Other phases of silica		
Stishovite	Extremely fine grained inclusions, concentrated in the lamellae of PDFs	None
Coesite	Inclusions of very fine-grained polycrystalline aggregates, within diaplectic quartz glass or quartz with n < 1.48 from shocked upper crustal rocks at impact structures	Inclusions of large single crystals within high pressure minerals (pyrope, diamond) of deep-seated metamorphic rocks or volcanic diatremes
Lechatelierite	Inclusions within impact glasses and tektites or individual "melt" particles within impact breccias or ejecta of impact structures	As fulgurites, within sand and, occasionally, sandstone

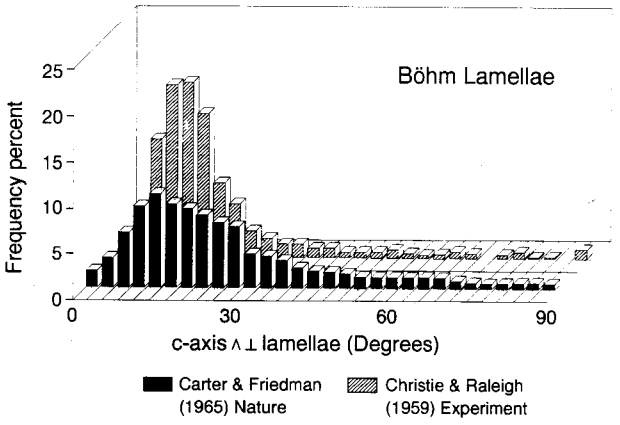


FIG. 31. Histogram of frequency percent of angle between c-axis and poles to Böhm lamellae, binned at 3°, in experimentally and tectonically deformed quartz. Note the lack of peaks at specific low Miller indices, which are distinctly different from frequency percent of indexed planes and angular distribution of planar microstructures produced by shock (*e.g.*, Figs. 11, 12, 22, 23, 24, 25, 26 and 27). Data are from Christie and Raleigh (1959) and Carter and Friedman (1965).

Two types of phase transformations are relevant to this discussion: (1) the occurrence of coesite and (2) the experimental amorphization of quartz by static compression. Coesite formed by static compression in deep-seated rocks has been discovered as inclusions in: (1) diamond recovered from volcanic diatremes (*e.g.*, Novgorodov *et al.*, 1990) and (2) high-pressure minerals, such as pyrope garnet from alpine and Caledonian fault zones (*e.g.*, Okay, 1993; Schertl *et al.*, 1993). In both cases, the crystal size of coesite, the textural setting and the paragenesis of coexisting minerals is distinctly different from shock-produced coesite, which is always extremely fine-grained and polycrystalline (Table 8).

Hemley *et al.* (1988) transformed α-quartz and coesite into amorphous phases on compression at 25 GPa–35 GPa in a diamond anvil cell at room temperature. Thus, one could argue that diaplectic quartz glass is no longer unequivocal evidence for shock compression. This interpretation, however, neglects not only fundamental differences in the formation mechanisms of diaplectic glass and statically produced amorphous quartz but also the textural and paragenetic setting. In naturally shocked rocks, diaplectic quartz glass coexists with coesite (and sometimes stishovite) and other minerals displaying characteristic shock features (*e.g.*, Chao, 1967, 1968; Robertson *et al.*, 1968; Stöffler, 1971a,b, 1972, 1984). As shown by TEM studies, the formation of diaplectic glass in shock events starts within PDFs and increases in abundance, with increasing pressure, until the entire crystal is amorphous. This is achieved at ~35 GPa, which is distinctly higher than the 25 GPa–30 GPa required for static high-pressure amorphization (Hemley *et al.*, 1988; Cordier *et al.*, 1993; Kingma *et al.*, 1993). Contrary to the view of Hemley *et al.* (1988), diaplectic glass forms by quenching of a high-pressure melt before the shock pressure is completely released (Langenhorst, 1994). Under static conditions, the amorphization, as suggested by Hemley *et al.* (1988), may indeed result from a direct solid-state transformation of α-quartz in the absence of melting. The term diaplectic glass should not be used for amorphous phases produced by static compression.

CONCLUDING REMARKS

As illustrated here and in Stöffler and Langenhorst (1994), quartz plays an important role in the recognition of terrestrial impact phenomena and as an experimentally calibrated shock barometer in shock metamorphism and cratering mechanical studies. Despite the number of studies on naturally and experimentally shocked quartz, there still remain several future directions for research. The following is a nonexclusive sample.

For the reasons noted in the Introduction, the identification of shocked quartz will continue to be one of the main tools in the recognition of terrestrial impact structures and lithologies. This contribution has illustrated, however, that systematic studies of shocked quartz at terrestrial impact structures are limited in number, particularly with respect to the parautochthonous lithologies of the floor of the structure. For example, we know of only one systematic study of shocked quartz at simple impact structures in crystalline targets and none at structures in sedimentary targets. Clearly, there is a case for additional systematic studies to enhance the observational database and confirm previous interpretations. Quartz is the principal shock barometer in the terrestrial environment, and additional studies will serve as constraints to future geomechanical models of cratering events.

Considerable advances have been made in the characterization and understanding of the formation and post-impact history of planar microstructures by recent TEM studies of experimentally and naturally shocked quartz. These studies, although ground breaking, have been generally exploratory in nature. There is a need for future systematic studies at terrestrial impact structures using such modern tools. Older shock recovery experiments and nuclear explosions, in which the recovered materials were examined only optically, also require reexamination by TEM to determine and document the nature of any basal (0001) planar microstructures. Shock recovery experiments at pressures <10 GPa, which is near the Hugoniot Elastic Limit of quartz, are almost completely lacking. It would also be very instructive to explore experimentally the pressure regime from ~3 GPa to 10 GPa, in order to extend the calibration of shock effects to the lowest possible pressure at which unequivocal shock effects are induced. This information would have implications for cratering mechanics. There is also a need for additional shock experiments on polycrystalline samples to address the differences identified in natural samples between the apparent behavior of quartz in non-porous and porous targets. Given the transition from non-decorated to decorated PDFs, it also appears that more experimental investigations of annealing behavior of naturally and experimentally shocked quartz and quartz-bearing rocks are in order. Experiments, such as these, are an attempt to duplicate more closely the complex character and history of natural impact events.

The discovery of shocked quartz in K/T boundary sediments has been highly relevant to settling the debate concerning the K/T boundary and has raised the standing of large-scale impact as an agent affecting terrestrial evolution in the larger geologic community. At present, no other events in the biostratigraphic record are unequivocally linked to equivalent impact events through the discovery of shocked quartz. Although the K/T impact event is currently a singular event in terms of impact-related mass extinctions, it was not a singular event in terms of the terrestrial cratering record. There must be evidence in the stratigraphic column of other major impact events resulting in dispersed ejecta, ranging from the "local" to global scale. At least in the case of major impacts on land, shocked quartz is an expected result. It would seem logical, therefore, to call for a systematic search for shocked quartz in the stratigraphic column to identify such events,

which could then be used as time-stratigraphic markers at various scales. We know of only one pioneering study of this type, which was performed on Devonian sediments (Schmitz *et al.*, 1994) and resulted in no discoveries of shocked quartz. There are relatively few impact events recognized currently in the stratigraphic column. Most are relatively young (<5 Ma) and, with the exception of the formation of the Acraman structure, all were initially identified as "anomalous" on the basis of other evidence (*e.g.*, the occurrence of microtektite bodies or geochemical anomalies; Grieve, 1996). Even in the case of Acraman, attention was drawn to the horizon because of the unusual occurrence of lithic volcanic clasts in an otherwise continuous section of shales. It is possible, therefore, that future discoveries will have a similar history, with the recognition of shocked quartz being secondary to the initial identification of a lithological oddity or geochemical anomaly.

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